# REVUB training dataset

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**Note**: This training dataset is periodically updated to be in line with the latest REVUB release. There may be lags between those releases and the updates of this training. The current version is updated to REVUB version 2.0.0.

Latest update: 4 November 2026

# Introduction

This exercise provides a hands-on practical introduction to working with the REVUB hydropower-VRE dispatch model. REVUB serves to examine the potential of individual hydropower plants with reservoir storage to support the "firming-up" of variable renewables (VRE), e.g. solar and wind power. This can be done by releasing more water from the reservoir during periods when VRE availability is low compared to demand, and retaining more water in the reservoir during periods when VRE availability is high.



In this fictional example, an international financial institution, the Planet Bank, has been approached by one of its member states, the Republic of Flexibar. Currently, the Republic's power mix is dependent on thermal power stations, but these are aging and may need to be decommissioned. The Republic's government is interested in finding financial support to **build a hydropower dam** at the site of the King Floribert Gorge.

The Bank has agreed to investigate the **feasibility and economic attractiveness** of this planned "King Floribert Hydropower Project", a proposed 1250-MW scheme that would impound a 8400-million $m^3$ storage reservoir (King Floribert Lake) on one of the country's principal rivers, requiring a 240-m high dam to be built in the Gorge.

Next to boosting the Republic's electricity generation, the project will also allow for a **large-scale irrigation project** to be developed downstream of the plant, receiving regulated water flow from the reservoir.

As part of its due diligence, and as part of its general interest in supporting the deployment of solar and wind power projects worldwide, the Planet Bank wants to know to what extent the King Floribert Hydropower Project would be able to provide storage and flexibility services to support VRE deployment in the Republic—in particular floating solar on King Floribert Lake, and wind power harvested on the mountain ridges surrounding the Gorge. This, they argue, would increase the attractiveness of financing the dam, and further compensate the planned decommissioning of thermal power plants.

In this exercise, you are the expert working for the Bank tasked with investigating this topic. Some of the principal questions are:

(i) How much **power generation** can we expect from the hydropower plant?
(ii) What would regular **drawdown-refill cycles** of the reservoir look like?

(iii) How would the hydropower plant perform under **extreme situations**, e.g. droughts?
(iv) How many MW of **solar and wind could be hybridised** with the project so that the hydro-VRE complex could effectively operate as a single unit?

You have been given **an initial dataset** on the King Floribert Hydropower Project's proposed design. This data includes river flow, precipitation, evaporation, downstream irrigation requirements, solar and wind resource strength, electricity demand, and technical data on the King Floribert Hydropower Plant design.

The REVUB model allows to run simulations from hourly-to-decadal timescale to investigate how to **optimise hydropower plant operation** in different contexts. It can provide insights on elements such as seasonal reservoir **drawdown-refilling cycles**, hour-to-hour **dispatch needs** of the hydro turbines to satisfy a demand with a given profile, and **operational strategies** under extreme events such as droughts.

The present (fictional) training dataset is meant to show how to run an appropriate REVUB simulation from start to end, starting with the raw data provided. This can inspire case studies based on real situations.

The training teaches important principles of **data initialisation**, **model calibration**, **scenario definition**, **simulation setup**, and **analysing results** at power plant and power mix level.

# Downloading the REVUB model

### GitHub repository

The REVUB model is provided as a collection of Python scripts, a control file, and input datasets compiled within the GitHub repository. **Clone the REVUB GitHub repository, including the training material, to a local folder on your system.**

The repository contains the following folders and files:

***1_REVUB_code/***
***2_Auxiliary_script/***
***3_Manual/***
***4_Training/***
***envs/***
***inputs/***
***outputs/***
***README***

Folder ***1_REVUB_code*** contains the REVUB Python code and control file. Folder ***2_Auxiliary_script*** contains an auxiliary script and control file. Folder ***3_Manual*** contains a full manual, folder ***4_Training*** hosts material for this training. The ***inputs*** folder stores the required input datasets, while the ***outputs*** folder stores the results generated by REVUB.

### Python environment

Anaconda (https://www.anaconda.com/download/success?reg=skipped) is used to manage the Python environment and the packages required by the MSR framework. The environment is defined in the *revub_env_update.yaml* file, located in the ***envs*** folder.

👉 Open an Anaconda Prompt or terminal, navigate to the ***envs*** folder and create and activate the environment using the following commands:

```
conda env create -f revub_env_update.yaml
conda activate revub_env_update
```

Spyder is used as integrated development environment (IDE) for running the Python scripts (launch *spyder* from the terminal inside the *revub_env_update* environment).

# Part 1: Data initialisation

### Task 1.1: Setting up the control file for simulation control

👉 Go to the ***1_REVUB_code*** folder and open the control file ***control_file.xlsx***. It has a pre-determined table structure. All basic **parameters for the simulations** and **visualisation of outputs** will be entered here.

👉 Go to the worksheet "*paths*".

This worksheet specifies the paths locating the input and the output folders. Specify your local paths to the ***inputs*** and ***outputs*** folders in the appropriate cells, as in **Figure 1**.

| | A | B | C |
|---|---|---|---|
| 1 | **parameter** | **path** | **notes** |
| 2 | path_inputs | C:\Users\adm_sesterl\OneDrive - Vrije Universiteit Brussel\Documenten\VUB Work Files\23006 REVUB tutorial\Training dataset\inputs | path to folder with input data sheets |
| 3 | path_outputs | C:\Users\adm_sesterl\OneDrive - Vrije Universiteit Brussel\Documenten\VUB Work Files\23006 REVUB tutorial\Training dataset\outputs | path where output will be located |

***Figure 1****: Entering file paths to input and output folders in the* ***control file*** → *paths worksheet.*

👉 Go to the worksheet "*Hydropower plant parameters*".

This sheet allows to specify all parameters listed in column A (*HPP_name*, *HPP_active*, etc.) for each separate hydropower plant that the user wishes to model.

These parameters are listed under different headers, starting with "*Hydropower plants*" (row 1), followed by "*Defining cascades*" (row 4), etc. Column B explains the meaning of the parameters.

Each column (from C onwards) will **contain data for one specific run** for an individual hydropower plant.

👉 Enter the name of the hydropower plant under investigation, e.g. "**King Floribert HPP**" or a similar name, in row 2 ("*HPP_name*") of column C, as in **Figure 2**.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | **1** | **0** | **0** |
| 4 | **Defining cascades [*Note: for advanced REVUB users; leave empty if unsure*]** | | | | |
| 5 | **HPP_cascade_upstream** | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant upstream of this one whose reservoir serves both. The upstream one must be present in this sheet, but can be set to HPP_active = 0. If it is set to HPP_active = 1, it will be forced to provide the outflow necessary to run the downstream one. | | | |
| 6 | **f_cascade_upstream** | ***[leave empty if unsure - default proportional to storage sizes will be used]*** In case this HPP is the principal flexibility provider in a cascade, and there is a hydropower plant upstream with reservoir that serves both plants, insert here the share of volume changes allocated to the upper reservoir. | | | |
| 7 | **HPP_cascade_downstream** | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant downstream of this one whose outflow it turbines. If the downstream one is present in this sheet and set to HPP_active = 1, it will be run with the outflow of this HPP as input data. | | | |
| 8 | **Importing temporal data** | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in m^3/s; hours in rows, years in columns) | | | |

*Figure 2: Each column in the* ***control file****'s sheet "Hydropower plant parameters" corresponds to one hydropower plant, where all parameters are entered.*

👉 The parameter "*HPP_active*" should be set to **1** for **King Floribert HPP**, and (for now) to **0** for the other columns to the right. In this way, any REVUB simulation will only use column C. See **Figure 2**.

👉 The rows 5-7 under header "*Defining cascades*" (row 4) can be left empty for now, as we are not yet asked to simulate a cascade.

👉 The rows 9-16 under header "*Importing temporal data*" (row 8) link each hydropower plant in the **control file** to the corresponding data in the ***data_(…).xlsx*** files in the ***inputs*** folder, which will **contain input time series**.

The name of the ***data_(…).xlsx*** file to which each of the rows 9-16 refers, is given by the variable name (in column A) minus "*HPP_name_*". **Thus, for instance, row 9 refers to the Excel file named *data_inflow.xlsx***. Below is the full set of Excel files in the ***inputs*** folder:

1. ***data_inflow.xlsx***. This file will be used to enter full hourly **time series of reservoir inflow**.

2. ***data_precipitation.xlsx***. This file will be used to enter full hourly **time series of precipitation gains** on the reservoir lake surface.

3. ***data_evaporation.xlsx***. This file will be used to enter full hourly **time series of evaporation losses** from the reservoir lake surface.

4. ***data_outflow_prescribed.xlsx***. This file will be used to enter full hourly **time series of required outflow** from the reservoir to meet downstream irrigation (and other) needs.

5. ***data_CF_solar.xlsx***. This file will be used to enter full hourly **time series of the capacity factor**[1] **of potential solar PV plants** built at the envisaged site(s).

6. ***data_CF_wind.xlsx***. This file will be used to enter full hourly **time series of the capacity factor of potential wind power plants** built at the envisaged site(s).

7. ***data_load.xlsx***. This file will be used to enter **full hourly time series of the electricity demand curve** that the hydropower plant (or, potentially, a hybridised hydro-VRE plant) will aim to meet; in other words, the production curve that the plant will aim to provide to the grid.

8. ***data_bathymetry.xlsx***. This file will contain data on the **shape of the reservoir** of the hydropower plant, so that REVUB can track changes in volume, area, and head in time.

These ***data_(…).xlsx*** files are currently still empty. In the next steps, we will collect the required data, fill the ***data_(…).xlsx*** files with that data, and then **link them to the control file**.

### Task 1.2: Setting up input data time series

All needed time series data needs to be entered in the empty Excel files in the following format: *all hours of the year* by row, and *all years of the simulation* by column.

Often, the limiting factor for the length that a simulation can span is the time spanned by **river flow data series**.

In this simple example, we have been given raw data on river inflow that spans a 26-year period (1980-2005). This data is provided in the repository folder ***4_Training_dataset*** in the file ***raw_data.xlsx*** → worksheet "*inflow*".

In many cases, input data would not be available at full hourly resolution across multiple years[2]. Rather, it is typical for the data to come in either of the **following two formats**:

[1] The capacity factor of any power plant is taken to refer to the electricity that it generated over a certain time period (e.g. hour, day, season, year...) divided by the power that would have been generated if the plant had run at full capacity over that entire period. Here, the relevant period is hourly.

[2] But whenever it is, that is great, as the data can then be directly entered in the ***data_(…).xlsx*** files and most of the following steps can be ignored.

1) **Monthly resolution**, either for a single "typical" year, or for multiple years.
   - This often applies to inflow, precipitation, and evaporation data sets.

2) **Hourly resolution** (24-h format) **for a "typical" day** for each month.
   - This may e.g. apply to solar PV and/or wind data; for instance, data downloaded from the Global Solar Atlas or Global Wind Atlas comes in this format.

To enable a REVUB simulation, we must first **convert such data to (pseudo)-hourly format** (8760 data points per year) before filling the ***data_(…).xlsx*** files.

This can be done with the **auxiliary code** ***rearrange_data_to_hourly.py***, available on the repository folder ***2_Auxiliary_script***.

This file uses the **data read-in template** named ***rearrange_data_template.xlsx***, which has two worksheets corresponding to data format 1 ("*monthly_series*") and data format 2 ("*daily_bymonth_series*"). The read-in template can be found in the same folder ***2_Auxiliary_script***.

The auxiliary code allows the required data flow, from raw (monthly-resolution) data to REVUB input. These necessary steps are:

- Pasting raw data into ***rearrange_data_template.xlsx***.
- Converting the data to REVUB format by running script ***rearrange_data_to_hourly.py***. The rearranged data will appear in a new sheet in ***rearrange_data_template.xlsx***.
- Transferring the rearranged data into a sheet in file ***data_(…).xlsx***.
- Entering that sheet's name in the correct cell in the **control file**.

***Working with data in format "1" (monthly averages)***

As you see in the file ***raw_data.xlsx*** → worksheet "*inflow*", the inflow data is only available in monthly format, but spans multiple years (1980-2005; cf. **Figure 3**).

We are thus dealing with data of format "1" above.

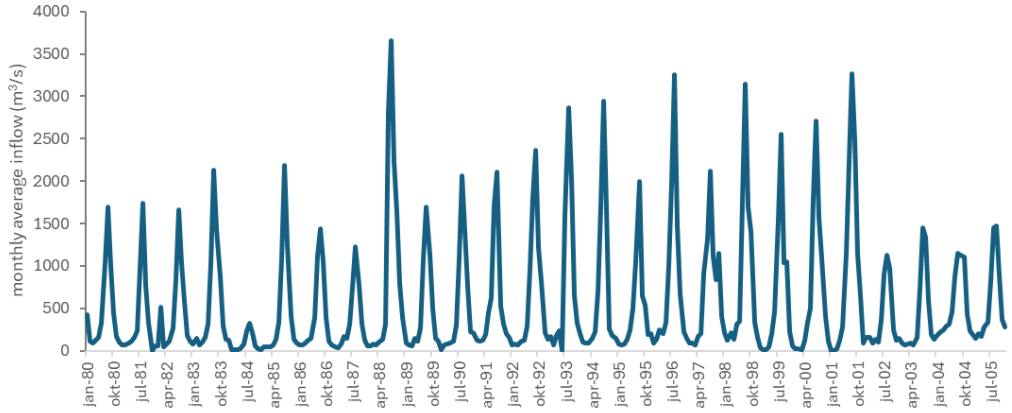



*Figure 3: The inflow data series at the site of King Floribert HPP.*

👉 We must use the read-in template ***rearrange_data_template.xlsx*** → worksheet "*monthly_series*" to convert the raw data into (pseudo)-hourly format.

- Here, we can **paste the raw data series** into one of the **dedicated columns**.
- We must also specify the **first and last year**, so that the **presence of leap years** is correctly accounted for.

👉 Thus, in ***rearrange_data_template.xlsx*** → worksheet "*monthly_series*", specify *first_year* = 1980 (row 3), *last_year* = 2005 (row 4), and copy/paste the monthly inflow data from the file ***raw_data.xlsx*** (312 values) into the rows below (row 5 and onwards), as in **Figure 4**.

👉 You need to also make sure the column is activated by specifying "1" in *activate for conversion* (row 2), and to avoid confusion with future data conversions, give it an appropriate *data series name* (row 1).

| | A | B | C | D |
|---|---|---|---|---|
| 1 | data series name | King Floribert HPP inflow (m3/s) | eries 2 (e.g. evaporation) | series 3 (e.g. precipitation) |
| 2 | activate for conversion (1 = yes, 0 = no) | 1 | 0 | 0 |
| 3 | first_year | 1980 | | |
| 4 | last_year | 2005 | | |
| 5 | *month 1* | 425 | | |
| 6 | *month 2* | 116 | | |
| 7 | *month 3* | 94 | | |
| 8 | *month 4* | 119 | | |
| 9 | *month 5* | 157 | | |
| 10 | *month 6* | 321 | | |
| 11 | *month 7* | raw data pasted 1038 | | |
| 12 | *month 8* | here 1692 | | |
| 13 | *month 9* | 990 | | |
| 14 | *month 10* | 439 | | |
| 15 | *month 11* | 171 | | |
| 16 | *month 12* | 97 | | |
| 17 | *month 13* | 72 | | |
| 18 | *month 14* | 73 | | |
| 19 | *month 15* | 96 | | |
| 20 | *month 16* | 118 | | |
| 21 | *month 17* | 157 | | |
| 22 | *month 18* | 230 | | |

*Figure 4: Pasting monthly raw data into the **rearrange_data_template.xlsx** file (worksheet "monthly_series") for conversion to (pseudo)-hourly data.*

### 🏃 CODE RUN 🏃

Now, we are ready to **carry out the format conversion**. To do this, run the Python script ***rearrange_data_to_hourly.py*** in the same folder as the template ***rearrange_data_template.xlsx***.

Running the file successfully will append a worksheet with the relevant data series name to ***rearrange_data_template.xlsx***. See **Figure 5** below for a visualisation.

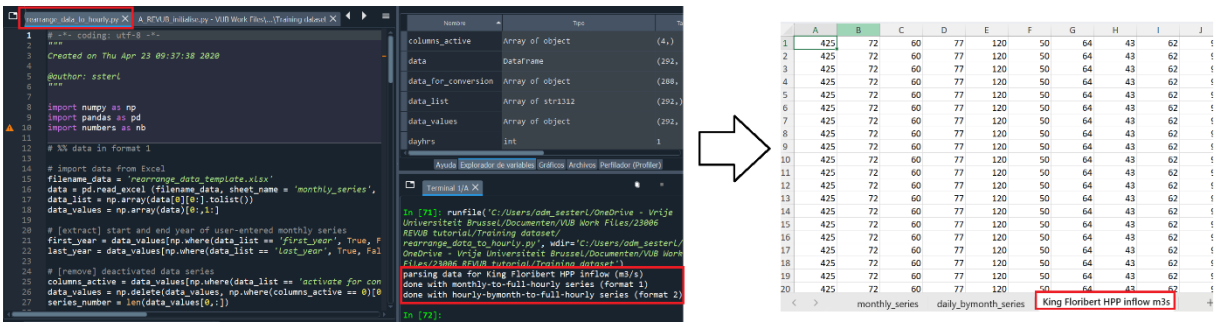



*Figure 5: The output of the **rearrange_data_to_hourly.py** code, converting monthly-resolution into (pseudo)-hourly-resolution data and transferring those to the template **rearrange_data_template.xlsx**.*

👉 **Copy the contents of that worksheet**, which contains the converted data series in the correct hourly format for REVUB, with hours in rows and years in columns, and **transfer them into a worksheet in the destination Excel file**, in this case ***data_inflow.xlsx***, as shown in **Figure 6**.

👉 The worksheet into which the data is transferred **requires a clear identifier** as name, e.g. "***King Floribert***" in **Figure 6**.

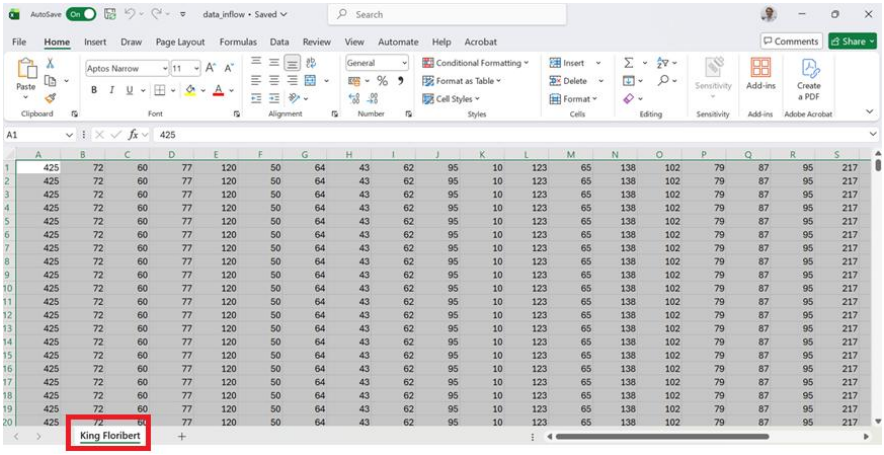



*Figure 6: Pasting hourly-resolution inflow data into an appropriately-named worksheet in the file **data_inflow.xlsx**.*

👉 You need to **input this identifier** in the relevant **row in the control file** (**Figure 7**); in the case of reservoir inflow, as in this example, it will be row 9 (parameter "*HPP_name_data_inflow*"). This links the hydropower plant we are simulating to its inflow data.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 8 | **Importing temporal data** | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in **m^3/s**; hours in rows, years in | King Floribert | | |
| 10 | **HPP_name_data_lateral_flow** | ***[leave empty if unsure - only in case of cascade calculations]*** name of corresponding worksheet in the "data" Excel sheets (use the "inflow" workbook) from which to pull data on lateral inflow to a downstream cascade plant (hourly flow in **m^3/s**; hours in rows, years in columns) | **name of the worksheet in "*data_inflow.xlsx*" to be read in** | | |
| 11 | **HPP_name_data_precipitation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (precipitation flux in **kg/m^2/s**; hours in rows, years in columns) | | | |

*Figure 7: Linking inflow data time series to the correct hydropower plant column in the **control file**.*

Next, follow the same principle to repeat the process for a few **other time series parameters** that come in similar raw formats.

👉 To fill ***data_precipitation.xlsx***: use the file ***raw_data.xlsx*** → worksheet "*precipitation*".

- This data is also given at monthly resolution for the period 1980-2005, and hence the **exact same procedure** can be applied as for the inflow (format 1).
- Note that you do not have to delete any previously pasted time series from the read-in template ***rearrange_data_template.xlsx***. You can simply **add columns to paste new data** (and **toggle the activate/disactivate option** in the second row).

👉 To fill ***data_evaporation.xlsx***: use the file ***raw_data.xlsx*** → worksheet "*evaporation*".

- For evaporation, unfortunately, you only have data for a "typical year" at your disposal. Thus, our best option is probably to assume that this data more or less holds for every year.
- The solution is thus to **repeat this time series 26 times first** (corresponding to 26 years in the period 1980-2005) and then **carry out the same procedure as above** for the inflow and precipitation.
- The time series repeated 26 times has already been prepared in the file ***raw_data.xlsx*** (data format 1).

***Working with data in format "2" (typical hourly curves by month)***

Now, have a look at the file ***raw_data.xlsx*** → worksheet *"outflow_prescribed"*. This sheet shows estimated **downstream irrigation requirements** for agricultural activities that will be developed downstream of the King Floribert hydropower plant.

The data are given **by hour of the day for each month of the year**: each month has **different irrigation needs**, in function of the crops' seasonal cycles and of rainfall; and **nighttime irrigation** is deemed more efficient than daytime irrigation.

👉 Since this data is given in format 2 (see above), we must parse the data into the read-in template ***rearrange_data_template.xlsx*** → worksheet "*daily_bymonth_series*" (as shown in **Figure 8**), and subsequently run the auxiliary script *rearrange_data_to_hourly.py*.

> 🏃 **CODE RUN** 🏃
>
> Run again the auxiliary code *rearrange_data_to_hourly.py*.

| | A | B | C |
|---|---|---|---|
| 1 | data series name | Irrigation needs downstream of King Floribert HPP | eries B |
| 2 | activate for conversion (1 = yes, 0 = no) | 1 | |
| 3 | first_year | 1980 | |
| 4 | last_year | 2005 | |
| 5 | Jan 0h | 100 | |
| 6 | Jan 1h | 100 | |
| 7 | Jan 2h | 100 | |
| 8 | Jan 3h | 100 | |
| 9 | Jan 4h | raw data pasted 100 | |
| 10 | Jan 5h | here 150 | |
| 11 | Jan 6h | 150 | |
| 12 | Jan 7h | 150 | |
| 13 | Jan 8h | 150 | |
| 14 | Jan 9h | 50 | |
| 15 | Jan 10h | 50 | |
| 16 | Jan 11h | 50 | |
| 17 | Jan 12h | 50 | |
| 18 | Jan 13h | 50 | |
| 19 | Jan 14h | 50 | |
| 20 | Jan 15h | 50 | |
| 21 | Jan 16h | 50 | |
| 22 | Jan 17h | 50 | |
| 23 | Jan 18h | 75 | |

*Figure 8: Pasting average hourly profiles by month into the **rearrange_data_template.xlsx** file (worksheet "daily_bymonth_series") for conversion to (pseudo)-hourly data.*

👉 Once the script is run, we can once again **copy the contents of the corresponding new worksheet** in ***rearrange_data_template.xlsx*** and **transfer** these into the file ***data_outflow_prescribed.xlsx***, and **link** the data to the **control file** as explained above.

After linking the required data files, the **control file** should look as in **Figure 9**. The user can freely choose the identifiers; in **Figure 9**, we have used "King Floribert" for all.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 8 | **Importing temporal data** | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in **m^3/s**; hours in rows, years in | King Floribert | | |
| 10 | **HPP_name_data_lateral_flow** | ***[leave empty if unsure - only in case of cascade calculations]*** name of corresponding worksheet in the "data" Excel sheets (use the "inflow" workbook) from which to pull data on lateral inflow to a downstream cascade plant (hourly flow in **m^3/s**; hours in rows, years in columns) | | | |
| 11 | **HPP_name_data_precipitation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (precipitation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | | |
| 12 | **HPP_name_data_evaporation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (evaporation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | | |
| 13 | **HPP_name_data_outflow_prescri** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data on prescribed (environmental/irrigation) outflow (hourly in **m^3/s**; hours in rows, years in columns) | King Floribert | | |
| 14 | **HPP_name_data_CF_solar** | ***[leave empty if scenario has no solar]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly solar CF as **fraction**/percentage; hours in rows, years in columns) | | | |
| 15 | **HPP_name_data_CF_wind** | ***[leave empty if scenario has no wind]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly wind CF as **fraction**/percentage; hours in rows, years in columns) | | | |
| 16 | **HPP_name_data_load** | ***[leave empty in calibration run]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly load **normalised** to a mean of unity) | | | |

*Figure 9: Linking the **data_XXX.xlsx** files to the corresponding hydropower plant in the **control file**.*

### Task 1.3: Setting up bathymetry series

We now turn to the data in the **control file** → sheet "*Hydropower plant parameters*", under header "*Importing bathymetry*" in row 17. Here, the idea is to link the **control file** with ***data_bathymetry.xlsx***, which needs to contain data on the **reservoir shape**.

Using geospatial analysis, engineers from the Republic of Flexibar have estimated this reservoir shape (**volume-area-head relationship**).

👉 The raw data for Lake Floribert's bathymetry is given in ***raw_data.xlsx*** → sheet "*bathymetry*". This data is to be pasted into ***data_bathymetry.xlsx*** as in **Figure 10**: volume in the first column, lake area in the second, hydraulic head in the third, and no headers [3,4].

| | A | B | C |
|---|---|---|---|
| 1 | 8.40E+09 | 2.00E+08 | 220.00 |
| 2 | 8.39E+09 | 2.00E+08 | 219.93 |
| 3 | 8.38E+09 | 2.00E+08 | 219.86 |
| 4 | 8.37E+09 | 2.00E+08 | 219.79 |
| 5 | 8.37E+09 | 1.99E+08 | 219.72 |
| 6 | 8.36E+09 | 1.99E+08 | 219.65 |
| 7 | 8.35E+09 | 1.99E+08 | 219.58 |
| 8 | 8.34E+09 | 1.99E+08 | 219.51 |
| 9 | 8.33E+09 | 1.99E+08 | 219.44 |
| 10 | 8.32E+09 | 1.99E+08 | 219.37 |
| 11 | 8.32E+09 | 1.99E+08 | 219.30 |
| 12 | 8.31E+09 | 1.98E+08 | 219.23 |
| 13 | 8.30E+09 | 1.98E+08 | 219.16 |
| 14 | 8.29E+09 | 1.98E+08 | 219.09 |
| 15 | 8.28E+09 | 1.98E+08 | 219.02 |
| 16 | 8.27E+09 | 1.98E+08 | 218.94 |
| 17 | 8.27E+09 | 1.98E+08 | 218.87 |
| 18 | 8.26E+09 | 1.98E+08 | 218.80 |
| 19 | 8.25E+09 | 1.98E+08 | 218.73 |
| 20 | 8.24E+09 | 1.97E+08 | 218.66 |

Lake Floribert

*Figure 10: Entering volume (first column), area (second column), and head (third column) in **data_bathymetry.xlsx**.*

[3] Note that the hydraulic head of the lake is of far higher importance to the eventual simulation outcomes than the lake area. This is because the power generated scales linearly with the hydraulic head, whereas the lake area is only important to calculate the correct total amount of lake surface evaporation and/or captured precipitation, which is usually a small fraction of the overall water budget of the reservoir.

[4] The more granular the bathymetric dataset, the more accurate the REVUB simulations will be in terms of reservoir dynamics. However, it is not strictly necessary to use a highly detailed (1000-point) dataset such as the one in this example. One could run a simulation with a fraction of that.

👉 Again, as above, the worksheet with the data in ***data_bathymetry.xlsx*** requires an identifier, which then needs to be reflected in the **control file**, as in **Figure 11**.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 17 | **Importing bathymetry** | | | | |
| 18 | **HPP_name_data_bathymetry** | name of corresponding worksheet in the "data" Excel sheets from which to pull data (first column: volume in **m^3**; second: surface area in **m^2**; third: head in | Lake Floribert | | |
| 19 | **Setting solar/wind mix** | | | | |
| 20 | **c_solar_relative** | ***[leave empty for scenarios without solar/wind]*** fraction of solar in the solar/wind portfolio | | | |

*Figure 11: Linking the correct bathymetry data sheet in **data_bathymetry.xlsx** to a hydropower plant in the **control file**.*

This data is needed to allow **calculating variations in hydraulic head drop and reservoir surface extent** as consequence of changes in stored volume.

### Task 1.4: Specifying technical and operational hydropower plant data

Now that we have set up the input time series and bathymetry, we need to specify **technical and operational parameters** of the hydropower plant.

These include the installed capacity of the plant (MW); the number of installed turbines; the maximum operationally achievable volume/surface/head (note that these may lay somewhere on the bathymetry curve from **Task 1.3** that is not necessarily the curve's endpoint), and others.

A full list of these parameters for the King Floribert hydropower plant is given in ***raw_data.xlsx*** → worksheet "*other technical parameters*".

👉 These data need to be **entered into the appropriate cells** of the **control file** → worksheet "*Hydropower plant parameters*", as in **Figure 12** below, under the header "*Setting technical hydropower plant parameters*" in row 21. In ***raw_data.xlsx***, they are given in the right order to **simply copy-and-paste** them all at once.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 21 | **Setting technical hydropower plant parameters** | | | | |
| 22 | **P_r_turb** | installed capacity in **MW** | 1250 | | |
| 23 | **V_max** | maximum lake volume in **m^3** | 8.4E+09 | | |
| 24 | **A_max** | maximum lake area in **m^2** | 2.0E+08 | | |
| 25 | **h_max** | maximum head in **m** | 220 | | |
| 26 | **Q_max_turb** | maximum discharge (total of all turbines) in **m^3/s** | 1000 | | |
| 27 | **no_turbines** | number of turbines (units) | 4 | | |
| 28 | **eta_turb** | turbine efficiency as fraction *[note: power factor can be included here]* | 0.80 | | |
| 29 | **f_initial_frac** | initial filling fraction of lake volume | 0.80 | | |
| 30 | **dP_ramp_turb** | turbine ramp rate in % of full capacity / min | 0.03 | | |
| 31 | **f_stop** | fraction of lake volume at which production stops | 0.10 | | |
| 32 | **f_restart** | fraction of lake volume at which production restarts after stopping | 0.20 | | |
| 33 | **f_opt** | optimal filling fraction f_opt (eq. S4, S5) | 0.80 | | |
| 34 | **f_spill** | fraction f_spill beyond which spilling starts (eq. S7) | 0.95 | | |
| 35 | **f_reg** | ***[leave empty if unsure - default determined by storage size will be used]*** which fraction of the incoming water is allocated for regulated use | | | |
| 36 | **min_load_turbine** | minimum load on one single turbine to ensure high-efficiency range (fraction) | 0.30 | | |
| 37 | **d_min** | ***[leave empty if unsure - default determined by minimum turbine load will be used]*** which fraction of the regulated use (two lines above) must be dispatched at stable level (eq. S4, S5) | | | |
| 38 | **alpha** | alpha (eq. S6) for conventional HPP operation rule curve (eq. S4) | 0.67 | | |
| 39 | **gamma_hydro** | gamma (eq. S4) for conventional HPP operation rule curve (eq. S4) | 10 | | |
| 40 | **mu** | mu parameter to control spilling (eq. S7) | 0.10 | | |
| 41 | **f_size** | this percentile controls the amount of allowed VRE overproduction (represents the % of time in which hydro+VRE may not exceed average ELCC) | 100 | | |

*Figure 12: Specifying technical and operational hydropower plant parameters in the **control file**.*

### Task 1.5: Setting simulation period

In the **control file**, worksheet "*General parameters*" contains a few more parameters that apply equally across all simulated hydropower plants (i.e. those with *HPP_active* = 1).

The parameters *year_start* and *year_end* are used to define the years we want to include in the simulation (i.e. the modelling period).

While we could use the full set of input data years (*year_start* = 1980; *year_end* = 2005), it is generally wise to start with a shorter time span to test quickly whether the simulation was set up correctly and gives sensible outputs.

👉 Start the initial test here with *year_start* = 1980 and *year_end* = 1983, i.e. using only the first four years.

👉 Also set the value *calibration_only* = 1. This allows to start with a light, generic simulation of drawdown-refill cycles of the reservoir, without more detailed dispatch calculations. **These drawdown-refill cycles will be used later as calibration in dispatch simulations, to ensure realistic lake level ranges**.

👉 The other parameters can be left as they are, as in **Figure 13** below.

After this step, a few parameters on the sheet "*Hydropower plant parameters*" will still be empty (e.g. *c_solar_relative* / *f_reg* / *year_calibration_start* / *f_init_BAL_start* / &c.).

Please leave them empty for now. Their use will become clear later.

| | A | B | C |
|---|---|---|---|
| 1 | **parameter** | **value** | **explanation** |
| 2 | | | |
| 3 | *simulation period* | | |
| 4 | year_start | 1980 | reference start year used in the simulation |
| 5 | year_end | 1983 | reference end year used in the simulation |
| 6 | years_skip | 0 | this parameter allows to skip the first *year_skip* years of the entered data (default: 0) |
| 7 | | | |
| 8 | *simulation settings* | | |
| 9 | calibration_only | 1 | select this option to not run hydro-solar-wind scenarios but only calibration run under regular hydropower operation (0 = no, 1 = yes) |
| 10 | option_storage | 0 | wish to activate pumped storage module STOR (cf. Manual, Note 6) for at least one hydropower plant or not? (0 = no, 1 = yes) |
| 11 | | | |
| 12 | *physical constants* | | |
| 13 | rho | 1000 | density of water (kg/m^3) (introduced in eq. S3) |
| 14 | g | 9.81 | gravitational acceleration (m/s^2) (introduced in eq. S8) |
| 15 | | | |
| 16 | *simulation accuracy* | | |
| 17 | N_ELCC | 1000 | This number defines the amount of discrete steps between 0 and *max(E_hydro + E_solar + E_wind)* |
| 18 | X_max | 2 | Number of loops for iterative estimation of *P_stable* in BAL/STOR (see eq. S9 & explanation below eq. S19). Usually 2 suffices; use 1 for quicker (but less accurate) simulations. |
| 19 | psi_min_threshold | 0 | When *min(Psi)* (eq. S21) is lower than this threshold, no further refinement loops are performed |
| 20 | | | |
| 21 | *miscellaneous simulation parameters* | | |
| 22 | T_fill_thres | 1 | represents number of years of filling as threshold for determining *f_reg* if left unspecified by user (see sheet "Hydropower plant parameters"). Default/recommended is unity, so the default *f_reg* represents the fraction of annual mean inflow that would take exactly one year to fill reservoir. |
| 23 | LOEE_allowed | 0 | allowed Loss of Energy Expectation as percentage of yearly ELCC |
| 24 | prevent_droughts_increase | 1 | select this option to discard potential solutions in which hydropower blackouts are worse under BAL/STOR than under CONV (0 = not prevented, 1 = prevented) [only during calibration period] |
| 25 | k_turb_saturation | 99 | this percentile is used to calculate the saturation rate of hydroturbine utilisation (introduced in eq. S28) |
| 26 | p_exceedance | 90 | this percentile is used to calculate the exceedance probability of delivered power (guaranteed capacity, MW). For P90, use 90; for P95, use 95, etc. |
| 27 | | | |

Paths | Hydropower plant parameters | General parameters | Plot

*Figure 13: Setting simulation period and establishing a calibration-only run in the **control file**.*

# Part 2: Calibration and dispatch runs

### Task 2.1: Setting up a calibration run (non-dispatch mode)

With your data sheets and **control file** set up, it is time to **run a first simulation** for the King Floribert hydropower plant.

👉 Open a console in your preferred environment (as said, here, we use Spyder) from which you can run the REVUB model.

> 🏃 **CODE RUN** 🏃
>
> First, run script *A_REVUB_initialise.py*. This script serves to read in all the data you parsed into their sheets in Part 1. In case any formatting errors occurred in this process, you will notice here.

If the code runs without error, your data have been **correctly read in**.

👉 Next, open the script ***B_REVUB_main_code.py***. It needs to be run from the same console; make sure the settings are appropriate for this (e.g. in Spyder, after opening the script, click *Run* → *Configuration per file* → *Run file with custom configuration* → check "*Execute in current console*" and "*Run in console's namespace instead of an empty one*").

> 🏃 **CODE RUN** 🏃
>
> Now run script ***B_REVUB_main_code.py***, the core code. You will see a progress counter in the console window to show how far along you are in the simulation, as in **Figure 14** below. **The simulation should take at most one minute. This is a calibration-run only; it is not yet a full dispatch simulation focused on matching the load curve.** The code only uses the hydrological balance and reservoir data to calculate a realistic drawdown-refill curve based on empirical equations. (For details, see the REVUB manual.)
>
> In the code and subsequent plots, the calibration result is referred to as "CONV" (from *conventional*).

```
HPP 1 / 1 : King Floribert HPP
> Simulating for period 1980 - 1983
 - using calibration period 1980 - 1983
(i) simulating CONV
f_reg = 0.56
Progress:  1 / 4  simulation years
Progress:  2 / 4  simulation years
Progress:  3 / 4  simulation years
Progress:  4 / 4  simulation years
> Note: Average spilling in CONV equal to 1.64 % of average
inflow.
done
> Calibration done; check outcomes and re-run with
calibration_only = 0 for full run
simulation finished
```

*Figure 14: Console output upon running script **B_REVUB_main_code.py** for a calibration run.*

Once you see the text "*simulation finished*" appear in the console (**Figure 14**), the simulation has **completed successfully**.

To test that we obtained sensible results, we will run a **quick visualisation** of the outcomes using the script ***C_REVUB_plotting_individual.py***, the plotting code for individual hydropower plant results.[5]

[5] There is also another script ***C_REVUB_plotting_multiple.py*** in which results for multiple hydropower plants included in the simulation (those with *HPP_active* = 1 in the **control file**) can be added together. We will use it later.

Before running the plotting script, we need to specify a few **plotting parameters** in the **control file**.

👉 In the **control file** → worksheet "*Plot power output (single HPP)*", we need to **specify the correct name** of the hydropower plant for which we want to plot results; it needs to match exactly the name in the worksheet "*Hydropower plant parameters*", which is done using a drop-down menu. See **Figure 15** for an example.

👉 We also need to turn on **which figures** we want to produce, and whether we want to produce output **tables**. For this calibration run, we only **activate the first three figures and the table**. All other fields can be left empty for now.

| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | **parameter** | **value** | **explanation** | | | | | |
| 2 | plot_HPP | King Floribert HPP | exact name of hydropower plant to plot | | | | | |
| 3 | plot_year | | for monthly and hourly graphs: choose which simulation year | | | | | |
| 4 | plot_month | | for hourly graph: choose month to plot (1 = Jan; 2 = Feb; etc.) | | | | | |
| 5 | plot_day_month | | for hourly graph: choose day of the month that the plot starts | | | | | |
| 6 | plot_num_days | | for hourly graph: number of days to show, starting at *plot_day* | | | | | |
| 7 | plot_RoR_part | | set to 1 to plot run-of-river component of electricity generatio | | | | | |
| 8 | plot_ELCC_line | | set to 1 to include a bold black line for ELCC, or to 0 to leave ou | | | | | |
| 9 | Figure1_on | 1 | set to 1 to plot bathymetry relations, or to 0 to turn off | | | | | |
| 10 | Figure2_on | 1 | set to 1 to plot hydraulic head and inflow/outflow IQ ranges, o | | | | | |
| 11 | Figure3_on | 1 | set to 1 to plot full time series of reservoir volume and inflow/ | | | | | |
| 12 | Figure4_on | | set to 1 to plot month-by-month power generation in year defi | | | | | |
| 13 | Figure5_on | | set to 1 to plot year-by-year power generation, or to 0 to turn | | | | | |
| 14 | Figure6_on | | set to 1 to plot hourly power generation from date *plot_day_n* | | | | | |
| 15 | Figure7_on | | set to 1 to plot regulated reservoir release statistics [see works | | | | | |
| 16 | Figure8_on | | set to 1 to plot hydroturbine use statistics, or to 0 to turn off | | | | | |
| 17 | Figure9_on | | set to 1 to plot operational regime statistics, or to 0 to turn off | | | | | |
| 18 | tables_on | 1 | set to 1 to output key time series in Excel format | | | | | |

*Figure 15: Specifying the right hydropower plant to plot output from a calibration run in the **control file**, worksheet "Plot power output (single HPP)".*

### 🏃 CODE RUN 🏃

Now run script ***C_REVUB_plotting_individual.py*** (do not forget to *Run in console's namespace instead of an empty one*). The script will produce three figures that will appear in the ***outputs*** folder, named "*[plant name]_Fig[X]*". It will also produce an Excel file with the name "*output_[plant name]_calibration.xlsx*".

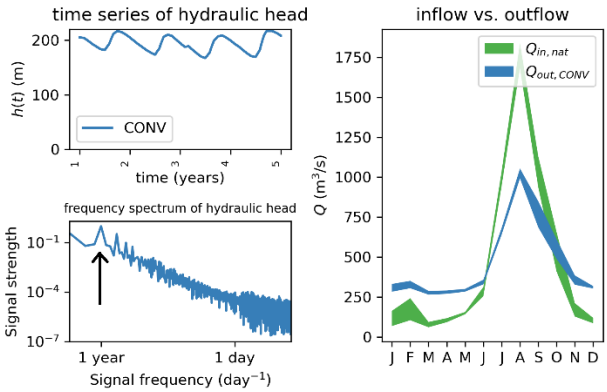
🎾 Examine the figures that have been produced:

The **first figure** simply reproduces the input bathymetry plots, for reference.

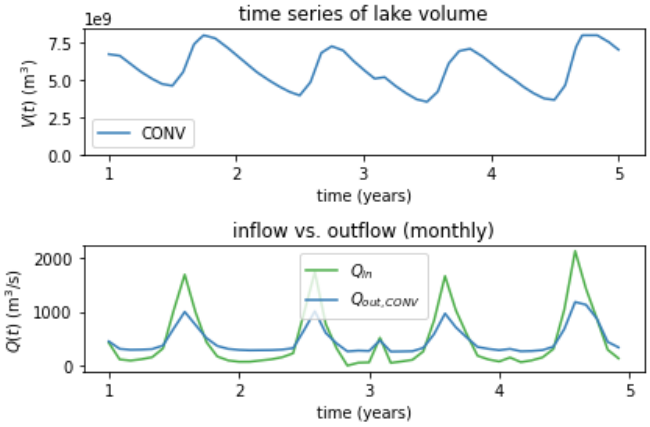
The **second figure** produces a time series of water levels, a frequency spectrum of water levels, and an average inflow-outflow plot. Here, we see reflected:

- The **yearly cycle** of the water levels (hydraulic head drop fluctuations; frequency spectrum with peak at 1 year, indicated by arrow below).

- The **regulation function** of the reservoir (inflow versus simulated outflow).




The **third figure** produces more detailed time series of lake filling level time series, and of inflow versus outflow for all simulated years.




This lake filling curve, whose corresponding **hourly time series** (volume and head) you can find in **the produced table**, will serve as **reference for our dispatch runs**, in other words: it represents a **rule curve**. Any dispatch run, with hourly-level decisions on hydropower plant dispatch, will have to **align with the rule curve** from the calibration run to the extent possible. REVUB optimises the operation in dispatch runs to ensure this.

> 💡 **TIP #1** 💡
>
> In case you are running a REVUB simulation **for an existing hydropower plant**, the rule curve from the calibration run can be **compared to observations** on actual water levels for additional calibration.
>
> In case the water level changes from the two curves do not match well—because one of the two has a much stronger drawdown-refill cycle than the other—the rule curve provided by REVUB can be calibrated to the observations by adapting the parameter *f_reg* (row 35 of the **control file** → sheet "*Hydropower plant parameters*").
>
> This parameter represents the **amount of water inflow allocated to storage** (and therefore, to regulated use). It need not be specified: REVUB can **estimate a "default" value** based on reservoir and inflow characteristics, which is given during the run (*f_reg* = 0.56 in **Figure 14**). However, this default value can be **over-ridden** by entering a different value for *f_reg* in the **control file** (**Figure 12**).
>
> In general: a **higher** *f_reg* **increases the amplitude** of the drawdown-refill cycle (more water is allocated for storage), a **lower** *f_reg* **decreases** it. In theory, *f_reg* = 0 corresponds to operating the plant as a run-of-river scheme.

### Task 2.2: Setting up a dispatch run (hydropower-only)

Let us now move from a calibration run (CONV) to a full dispatch run (BAL, for "*balancing*"). We will now set up the additional data needed to do a full dispatch run. We will assume that hydropower is dispatched flexibly, within its restrictions (ramp rates, minimum turbine loads, environmental and irrigation outflow, etc.) to **meet a target load curve**.

First, we need to **define that load curve** at hourly level.

Typically, such load data should be available in full hourly format (8760 data points per year), but in this instance, the electric utility of the Republic of Flexibar does not make full hourly data available and provided only a typical curve for each month. This data is provided in the file ***raw_data.xlsx*** → worksheet "*load*" (electricity demand).

The data has been normalised (average = 1; see **Figure 16**). The appropriate absolute level of the load curve (in MW) that the hydropower plant can meet if appropriately dispatched, will be calculated later by REVUB.

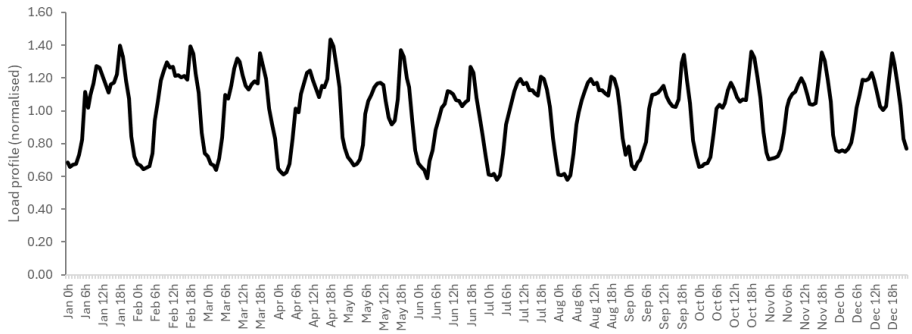



*Figure 16: The normalised load data series of the Republic of Flexibar.*

To fill ***data_load.xlsx***, use what you learnt in **Task 1.2**. Similar to the prescribed irrigation outflow, this data is given in format "2".

👉 **Parse the load data** into the template ***rearrange_data_template.xlsx*** → worksheet "*daily_bymonth_series*", and subsequently run the auxiliary script ***rearrange_data_to_hourly.py***, as before.

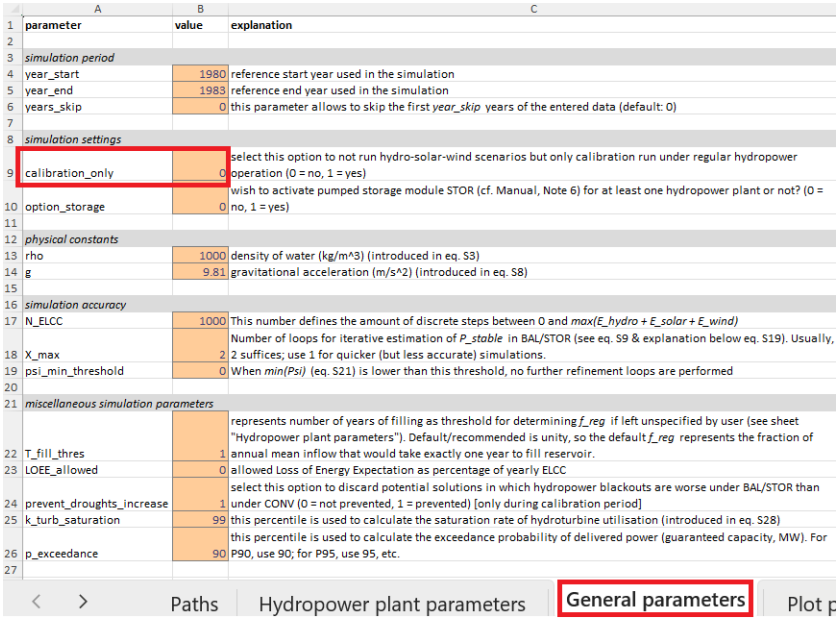
👉 **Copy the output array** and paste it in the file ***data_load.xlsx***.

👉 **Link the worksheet** to the **control file** using the same procedure as in **Task 1.2**; see **Figure 17**.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***[name 2]*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 8 | **Importing temporal data** | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in **m^3/s**; hours in rows, years in | King Floribert | | |
| 10 | **HPP_name_data_lateral_flow** | ***[leave empty if unsure - only in case of cascade calculations]*** name of corresponding worksheet in the "data" Excel sheets (use the "inflow" workbook) from which to pull data on lateral inflow to a downstream cascade plant (hourly flow in **m^3/s**; hours in rows, years in columns) | | | |
| 11 | **HPP_name_data_precipitation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (precipitation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | | |
| 12 | **HPP_name_data_evaporation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (evaporation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | | |
| 13 | **HPP_name_data_outflow_prescri** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data on prescribed (environmental/irrigation) outflow (hourly in **m^3/s**; hours in rows, years in columns) | King Floribert | | |
| 14 | **HPP_name_data_CF_solar** | ***[leave empty if scenario has no solar]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly solar CF as **fraction**/percentage; hours in rows, years in columns) | | | |
| 15 | **HPP_name_data_CF_wind** | ***[leave empty if scenario has no wind]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly wind CF as **fraction**/percentage; hours in rows, years in columns) | | | |
| 16 | **HPP_name_data_load** | ***[leave empty in calibration run]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly load **normalised** to a mean of unity) | Flexibar | | |

*Figure 17: Linking load data to the hydropower plant for a dispatch run.*

👉 **Go back** to the **control file** → sheet "*General parameters*". Set the value *calibration_only* = 0, as in **Figure 18**. Leave everything else unchanged.



| | parameter | value | explanation |
|---|---|---|---|
| 3 | *simulation period* | | |
| 4 | year_start | 1980 | reference start year used in the simulation |
| 5 | year_end | 1983 | reference end year used in the simulation |
| 6 | years_skip | 0 | this parameter allows to skip the first *year_skip* years of the entered data (default: 0) |
| 8 | *simulation settings* | | |
| 9 | calibration_only | 0 | select this option to not run hydro-solar-wind scenarios but only calibration run under regular hydropower operation (0 = no, 1 = yes) |
| 10 | option_storage | 0 | wish to activate pumped storage module STOR (cf. Manual, Note 6) for at least one hydropower plant or not? (0 = no, 1 = yes) |
| 12 | *physical constants* | | |
| 13 | rho | 1000 | density of water (kg/m^3) (introduced in eq. S3) |
| 14 | g | 9.81 | gravitational acceleration (m/s^2) (introduced in eq. S8) |
| 16 | *simulation accuracy* | | |
| 17 | N_ELCC | 1000 | This number defines the amount of discrete steps between 0 and *max(E_hydro + E_solar + E_wind)* |
| 18 | X_max | 2 | Number of loops for iterative estimation of *P_stable* in BAL/STOR (see eq. S9 & explanation below eq. S19). Usually, 2 suffices; use 1 for quicker (but less accurate) simulations. |
| 19 | psi_min_threshold | 0 | When *min(Psi)* (eq. S21) is lower than this threshold, no further refinement loops are performed |
| 21 | *miscellaneous simulation parameters* | | |
| 22 | T_fill_thres | 1 | represents number of years of filling as threshold for determining *f_reg* if left unspecified by user (see sheet "Hydropower plant parameters"). Default/recommended is unity, so the default *f_reg* represents the fraction of annual mean inflow that would take exactly one year to fill reservoir. |
| 23 | LOEE_allowed | 0 | allowed Loss of Energy Expectation as percentage of yearly ELCC |
| 24 | prevent_droughts_increase | 1 | select this option to discard potential solutions in which hydropower blackouts are worse under BAL/STOR than under CONV (0 = not prevented, 1 = prevented) [only during calibration period] |
| 25 | k_turb_saturation | 99 | this percentile is used to calculate the saturation rate of hydroturbine utilisation (introduced in eq. S28) |
| 26 | p_exceedance | 90 | this percentile is used to calculate the exceedance probability of delivered power (guaranteed capacity, MW). For P90, use 90; for P95, use 95, etc. |

Paths | Hydropower plant parameters | **General parameters** | Plot p

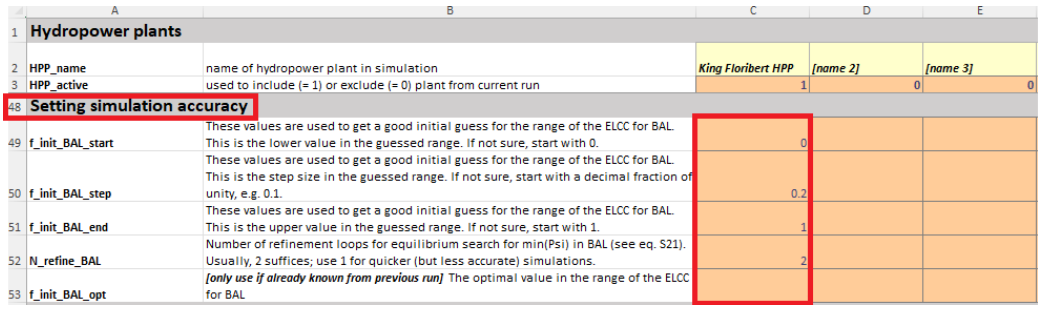
*Figure 18: Changing from calibration run to dispatch run.*

Last, we need to set parameters related to the **accuracy of the optimisation** in the **control file** → worksheet "*Hydropower plant parameters*", under the header "*Setting simulation accuracy*" in row 48.

The parameters *f_init_BAL_start*, *f_init_BAL_step*, and *f_init_BAL_end* (rows 49-51) provide a range with a step size. They are basically multipliers for a base value of a possible demand to be met (see also **TIP #2**).

REVUB will **loop across different levels of this demand**, from low to high using the specified range, and **identify an "optimum" where the highest demand is met** while still respecting all boundary conditions (such as respecting the rule curve from the calibration run). REVUB can then zoom in around this optimum at higher resolution and re-search more precisely, repeating this process *N_refine_BAL* times (row 52).

👉 The values as shown in **Figure 19** below permit a relatively quick simulation. Please use those as input: *f_init_BAL_start* = 0, *f_init_BAL_step* = 0.2, *f_init_BAL_end* = 1, *N_refine_BAL* = 2.



| | Hydropower plants | | | | |
|---|---|---|---|---|---|
| 2 | HPP_name | name of hydropower plant in simulation | *King Floribert HPP* | *[name 2]* | *[name 3]* |
| 3 | HPP_active | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 48 | **Setting simulation accuracy** | | | | |
| 49 | f_init_BAL_start | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the lower value in the guessed range. If not sure, start with 0. | 0 | | |
| 50 | f_init_BAL_step | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the step size in the guessed range. If not sure, start with a decimal fraction of unity, e.g. 0.1. | 0.2 | | |
| 51 | f_init_BAL_end | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the upper value in the guessed range. If not sure, start with 1. | 1 | | |
| 52 | N_refine_BAL | Number of refinement loops for equilibrium search for min(Psi) in BAL (see eq. S21). Usually, 2 suffices; use 1 for quicker (but less accurate) simulations. | 2 | | |
| 53 | f_init_BAL_opt | ***[only use if already known from previous run]*** The optimal value in the range of the ELCC for BAL | | | |

*Figure 19: Setting simulation accuracy parameters*

We are now ready for our **first dispatch run**.

> 🏃 **CODE RUN** 🏃
>
> Run script ***A_REVUB_initialise.py*** and then ***B_REVUB_main_code.py***. It should take 2-3 minutes.

You will see that it takes a little longer compared to the calibration-only run, as a full dispatch simulation has been added. After repeating the calibration, the code **cycles through different "refinement steps"** to find the optimal solution (**Figure 20**).




*Figure 20: Console output upon running script B_REVUB_main_code.py for a dispatch run.*

Before visualising the results, we need to **specify what we want to visualise** in the **control file**.

👉🏽 In the **control file** → worksheet "*Plot power output (single HPP)*", specify the additional parameters as follows **to activate figures relevant for a dispatch run**, for instance as in **Figure 21**.

The parameters *plot_year*, *plot_month*, *plot_day_month* and *plot_num_days* can be **freely chosen** by the user. They define the year (*plot_year*) for which the seasonal dispatch will be displayed, and the month (*plot_month*) for which the hourly dispatch will be displayed, starting on the day defined by *plot_day_month* and for a number of days equal to *plot_num_days*. Here, *plot_year* can be a number between 1 and 4, since we are running 4 simulation years; *plot_month* is a number between 1 and 12 for the months of the year; and *plot_day_month* is a number between 1 and 31 for the days of each month.

| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | **parameter** | **value** | **explanation** | | | | | |
| 2 | plot_HPP | King Floribert HPP | exact name of hydropower plant to plot | | | | | |
| 3 | plot_year | 2 | for monthly and hourly graphs: choose which simulation year | | | | | |
| 4 | plot_month | 4 | for hourly graph: choose month to plot (1 = Jan; 2 = Feb; etc.) | | | | | |
| 5 | plot_day_month | 12 | for hourly graph: choose day of the month that the plot starts | | | | | |
| 6 | plot_num_days | 3 | for hourly graph: number of days to show, starting at *plot_da* | | | | | |
| 7 | plot_RoR_part | 1 | set to 1 to plot run-of-river component of electricity generatio | | | | | |
| 8 | plot_ELCC_line | 1 | set to 1 to include a bold black line for ELCC, or to 0 to leave ou | | | | | |
| 9 | Figure1_on | 1 | set to 1 to plot bathymetry relations, or to 0 to turn off | | | | | |
| 10 | Figure2_on | 1 | set to 1 to plot hydraulic head and inflow/outflow IQ ranges, o | | | | | |
| 11 | Figure3_on | 1 | set to 1 to plot full time series of reservoir volume and inflow/ | | | | | |
| 12 | Figure4_on | 1 | set to 1 to plot month-by-month power generation in year defi | | | | | |
| 13 | Figure5_on | 1 | set to 1 to plot year-by-year power generation, or to 0 to turn | | | | | |
| 14 | Figure6_on | 1 | set to 1 to plot hourly power generation from date *plot_day_n* | | | | | |
| 15 | Figure7_on | 0 | set to 1 to plot regulated reservoir release statistics [see works | | | | | |
| 16 | Figure8_on | 0 | set to 1 to plot hydroturbine use statistics, or to 0 to turn off | | | | | |
| 17 | Figure9_on | 0 | set to 1 to plot operational regime statistics, or to 0 to turn off | | | | | |
| 18 | tables_on | 1 | set to 1 to output key time series in Excel format | | | | | |

*Figure 21: Adapting visualisation parameters in the **control file**, worksheet "Plot power output (single HPP)", to visualise results of a dispatch run.*

### 🏃🏽 CODE RUN 🏃🏽

Now run script ***C_REVUB_plotting_individual.py***. The script will produce six figures, and an additional table with dispatch run results named "*output_[plant name]_dispatch.xlsx*".

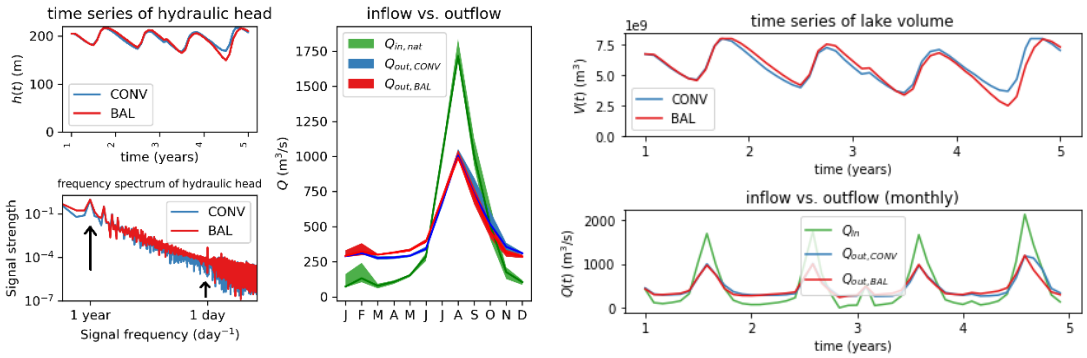
🎾 Examine again the figures that have been produced.

In the **second** and **third** figure, compared to the calibration run, an additional curve "BAL" has been added to the graphs. This curve corresponds to the dispatch run.

We see, among other things:

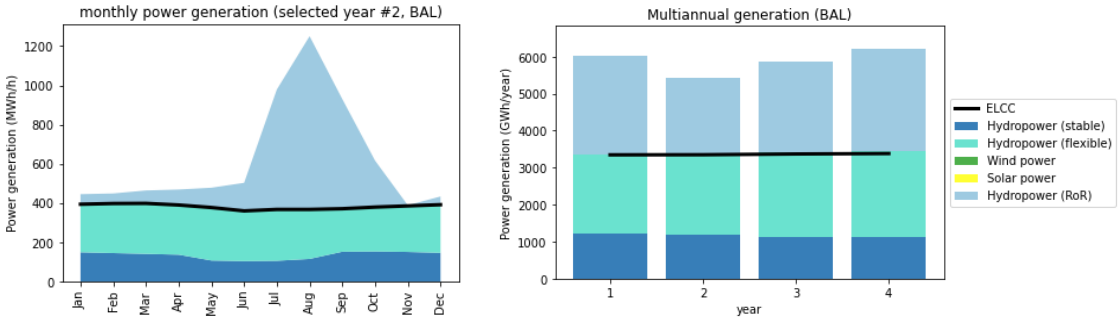
- The time series of water levels (expressed as hydraulic head) from the dispatch run (BAL) is **extremely close to the rule curve** from the calibration run (CONV). This is a result of the optimisation procedure of REVUB, finding the solution best aligned with the calibration run in terms of drawdown-refill cycles.

- The frequency spectrum of the dispatch run shows a peak at a frequency of **1 day** (see arrows). This reflects how hydropower has to **ramp up daily** to meet a given daily peak load (see **Figure 16**).



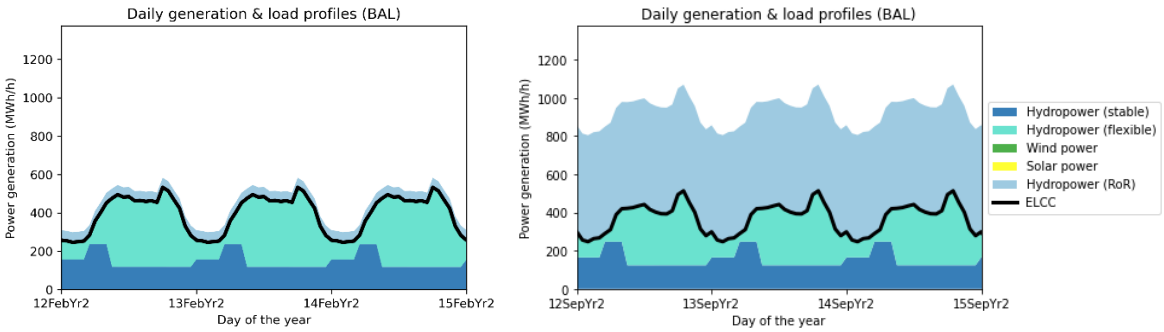

In the **fourth, fifth,** and **sixth** figures, we see the hydropower plant output at seasonal, annual, and hourly scales.

- At seasonal scale, we see that the reservoir is able to **contain part of the inflow**, but that the discharge peak in the months July-September is usually **too much to be stored**, and has to be released directly as "run-of-river" component (**fourth** and **fifth** figure).




- At hourly scale (**sixth** figure), we see how hydropower **ramps up during daytime**, especially evening hours, and **down at night-time**. The "stable" hydropower component reflects the minimum turbine load, as well as irrigation needs downstream, and cannot adapt to demand.[6]

[6] It can be seen that the irrigation component (cf. the curve used in **Task 1.2**) is higher than the minimum load component during the night and morning hours, but not during the rest of the day when the minimum load component automatically covers the irrigation requirements.




The line called "ELCC" represents the "**Effective Load Carrying Capability**". It refers to the **guaranteed load that the hydropower plant can meet** from hour to hour. The run-of-river component is not counted as contributing to this, since it cannot be stored and controlled.

### Task 2.3: The importance of the calibration period

Let us go back to the inflow data we are using (**Figure 22**). Clearly, the year 1984, directly following our model period so far (1980-1983), was **extremely dry**, with an **anomalous flow deficit** on the river.

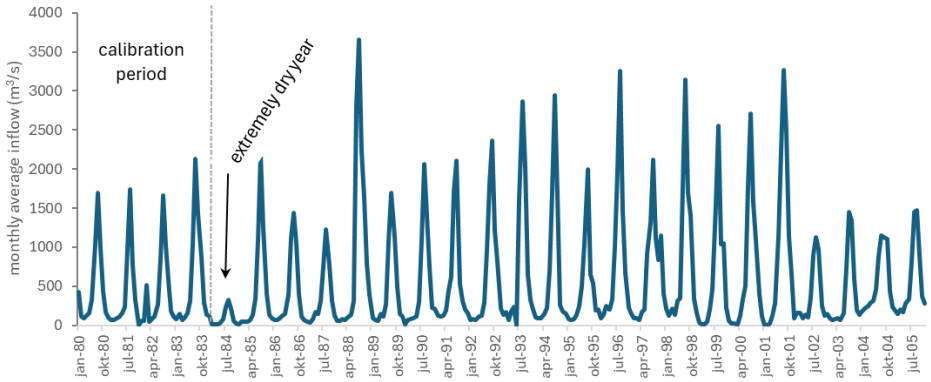



*Figure 22: The inflow data series at the site of King Floribert HPP, now with the calibration period indicated.*

Would the **operational principles** designed for the period 1980-1983 still work even under such extremely dry conditions?

We can investigate this by **keeping the same *calibration period***, but **changing the *model period***.

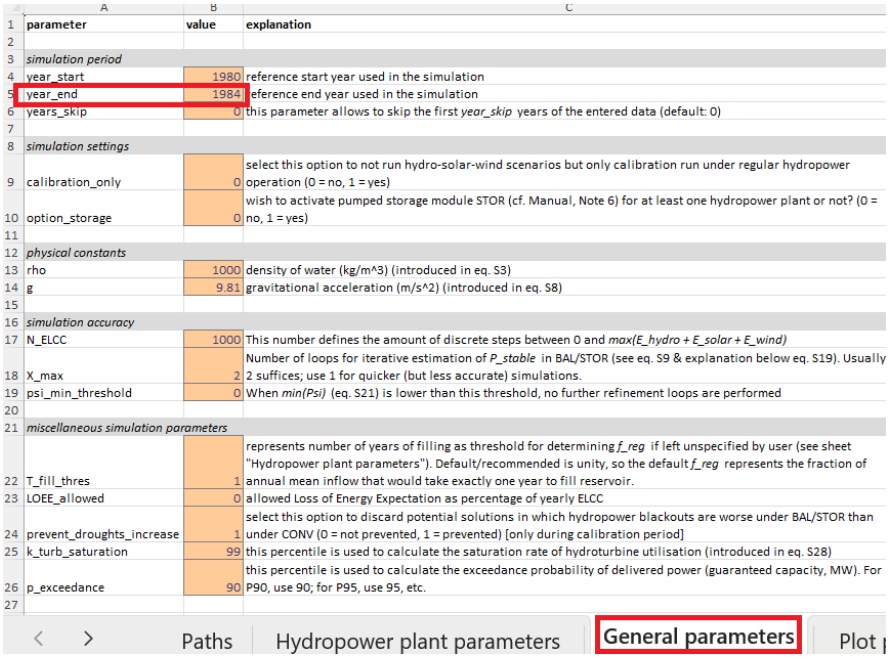
👉 Set the calibration period explicitly to 1980-1983 in the **control file** → worksheet "*Hydropower plant parameters*", by specifying, under the header "*Setting calibration period*" in row 42, the parameters *year_calibration_start* = 1980 and *year_calibration_end* = 1983 (equal to the modelling period so far), as in **Figure 23**.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | *King Floribert HPP* | *[name 2]* | *[name 3]* |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 1 | 0 | 0 |
| 42 | **Setting calibration period** | ***[leave empty if no preference - full period will be used]*** | | | |
| 43 | **year_calibration_start** | the first year of the multiannual period whose discharge average is to be used for the calibration of the conventional operation | 1980 | | |
| 44 | **year_calibration_end** | the last year of the multiannual period whose discharge average is to be used for the calibration of the conventional operation | 1983 | | |

*Figure 23: Specifying the calibration period explicitly.*

👉🏽 Now, set the modelling period to 1980-**1984** instead of 1980-**1983** on the sheet "*General parameters*" (**Figure 24**).

In this way, the model will run based on operating rules optimised for 1980-1983, and then run "blindly" into the dry year 1984, as if it were a regular year.



| | A | B | C |
|---|---|---|---|
| 1 | **parameter** | **value** | **explanation** |
| 2 | | | |
| 3 | *simulation period* | | |
| 4 | year_start | 1980 | reference start year used in the simulation |
| 5 | year_end | 1984 | eference end year used in the simulation |
| 6 | years_skip | 0 | this parameter allows to skip the first *year_skip* years of the entered data (default: 0) |
| 7 | | | |
| 8 | *simulation settings* | | |
| 9 | calibration_only | 0 | select this option to not run hydro-solar-wind scenarios but only calibration run under regular hydropower operation (0 = no, 1 = yes) |
| 10 | option_storage | 0 | wish to activate pumped storage module STOR (cf. Manual, Note 6) for at least one hydropower plant or not? (0 = no, 1 = yes) |
| 11 | | | |
| 12 | *physical constants* | | |
| 13 | rho | 1000 | density of water (kg/m^3) (introduced in eq. S3) |
| 14 | g | 9.81 | gravitational acceleration (m/s^2) (introduced in eq. S8) |
| 15 | | | |
| 16 | *simulation accuracy* | | |
| 17 | N_ELCC | 1000 | This number defines the amount of discrete steps between 0 and *max(E_hydro + E_solar + E_wind)* |
| 18 | X_max | 2 | Number of loops for iterative estimation of *P_stable* in BAL/STOR (see eq. S9 & explanation below eq. S19). Usually 2 suffices; use 1 for quicker (but less accurate) simulations. |
| 19 | psi_min_threshold | 0 | When *min(Psi)* (eq. S21) is lower than this threshold, no further refinement loops are performed |
| 20 | | | |
| 21 | *miscellaneous simulation parameters* | | |
| 22 | T_fill_thres | 1 | represents number of years of filling as threshold for determining *f_reg* if left unspecified by user (see sheet "Hydropower plant parameters"). Default/recommended is unity, so the default *f_reg* represents the fraction of annual mean inflow that would take exactly one year to fill reservoir. |
| 23 | LOEE_allowed | 0 | allowed Loss of Energy Expectation as percentage of yearly ELCC |
| 24 | prevent_droughts_increase | 1 | select this option to discard potential solutions in which hydropower blackouts are worse under BAL/STOR than under CONV (0 = not prevented, 1 = prevented) [only during calibration period] |
| 25 | k_turb_saturation | 99 | this percentile is used to calculate the saturation rate of hydroturbine utilisation (introduced in eq. S28) |
| 26 | p_exceedance | 90 | this percentile is used to calculate the exceedance probability of delivered power (guaranteed capacity, MW). For P90, use 90; for P95, use 95, etc. |
| 27 | | | |



*Figure 24: Adapting the modelling period to extend beyond the calibration period.*

### 🤾🏽 CODE RUN 🤾🏽

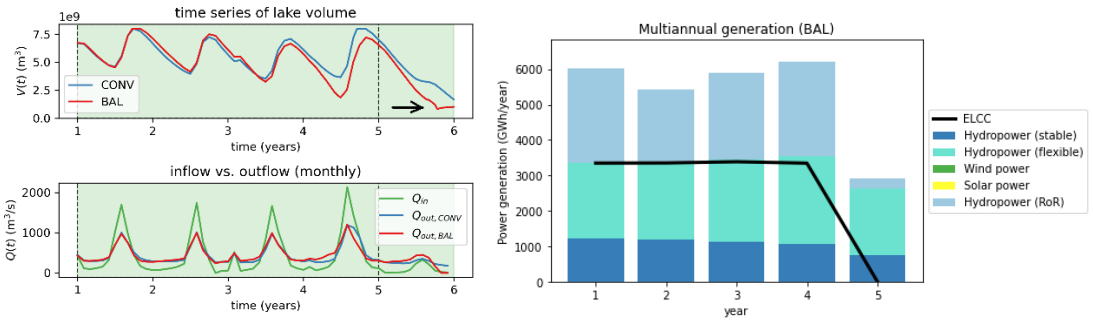
Now run all three REVUB scripts, as before (***A_REVUB_initialise.py***, ***B_REVUB_main_code.py*** and ***C_REVUB_plotting_individual.py***).

🎾 Examine what happens to lake levels and electricity generation.

- The time series of lake volume (third produced figure) shows a strong **drop of lake levels** in the final simulation year, including what looks like a **shutdown of the plant** to let lake levels slowly recover (indicated with the black arrow in the below figure).
    - For clarity, since the calibration does not span the full model period, the calibration period will be automatically indicated with vertical dashed lines (1980-1983) whereas the model period is indicated with a green shaded area (1980-1984).

- The multiannual generation (fifth produced figure) shows how **hydropower generation drops** and the **regular ELCC cannot be guaranteed** throughout the final simulation year.

- Clearly, the dry year causes lake levels drop so low that hydropower generation cannot be maintained, and **hydropower generation is curtailed** towards the end of the model period.




How could we have prevented this "drought crisis"? Clearly, the operational rules designed on the basis of 1980-1983 were too optimistic: they did not account for the possibility of an extremely dry year.

👉🏽 Check what would happen if we took this possibility into account. Change the calibration period from 1980-**1983** to 1980-**1984** (using *year_calibration_end* = 1984; see **Figure 23**).

> 🏃 **CODE RUN** 🏃
>
> Now run REVUB again.

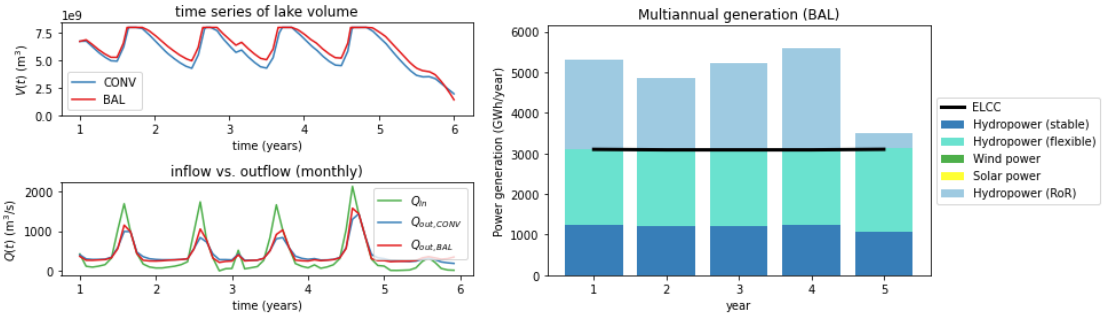
🎾 Examine the differences in lake level and power generation outcomes, compared to the previous run.

- We see that lake levels remain higher throughout, and the ELCC is maintained year-round, even towards the end of the final simulation year.

- With the new calibration period including a very dry year, the possibility of a massive drought is *a priori* taken into account, and the REVUB algorithm optimises operational rules to prevent hydropower plant shutdowns.

- The result is **more prudent reservoir management**. More water is stored in the reservoir in normal years, instead of being released.

- As a result, some of the hydropower generation is **sacrificed in normal years**, but the **drought impact is mitigated**.



💡 **TIP #2** 💡

The ranges of *f_init_BAL* given in **Figure 19** represent a proxy for the ELCC compared to the firm hydropower capacity. REVUB loops through different values and tries to find the highest ELCC possible within the specified technical and environmental limitations, i.e. the optimal ELCC.

It is useful to note down the optimal value *f_opt_BAL* (which will appear in the variable explorer) that a simulation finds (**Figure 20**) in case you need to re-run it later. It would be a waste of time to re-run the full optimisation every time you repeat this exact scenario. You can simply instruct REVUB to run directly at the optimal parameterisation.

This can be done by setting the parameter *f_init_BAL_opt* (row 53 in the **control file** → sheet "*Hydropower plant parameters*") which will supersede the initial range and force REVUB to check only one single operational scheme: aiming for the ELCC we know is optimal for this particular combination of scenario parameters.

For instance, for the case in **Figure 20**, we would set *f_init_BAL_opt* = *f_opt_BAL* = 0.54.

In **Task 3.3**, you will use this tip.

### Task 2.4: Setting up a dispatch run (hydropower with solar and wind)

We will now examine how the dispatch would change if we **included solar and wind** in the mix. The idea is that solar and wind will meet part of the load. Therefore, **hydropower dispatch will change** as well.

With the correct dispatch strategy, we can **maximise the amount of solar and wind** that can be integrated into a load-following mix of different sources without deviating too much from the rule curve.

First, the **solar and wind profiles** need to be defined.

👉🏽 To fill ***data_CF_solar.xlsx*** and ***data_CF_wind.xlsx***: use the file ***raw_data.xlsx*** → worksheet *"solar power"* and "*wind power*". This sheet provides estimates of the hourly capacity factor of solar panels and wind turbines (**Figure 25** and **Figure 26**) near the King Floribert project for each month of the year, i.e. format "2". Use the auxiliary code ***rearrange_data_to_hourly.py***.

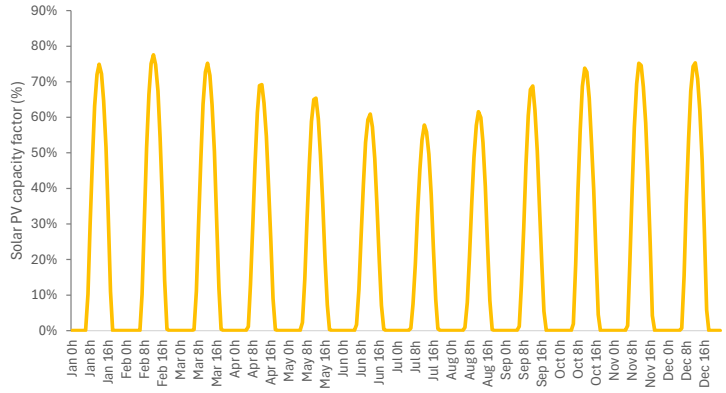



*Figure 25: The raw data provided on solar power potential at daily and seasonal scale.*

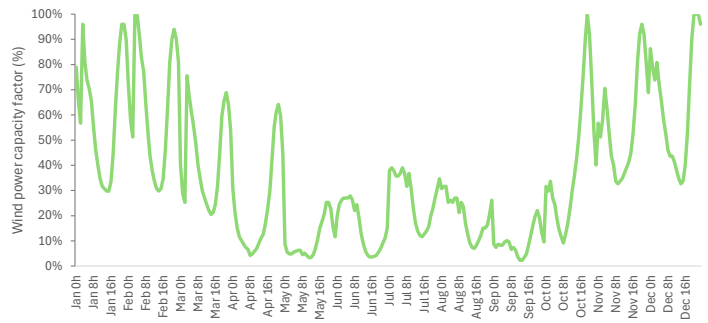



*Figure 26: The raw data provided on solar power potential at daily and seasonal scale.*

👉 We do not want to lose the settings from our previous (hydropower-only) dispatch run. Therefore, **create a new scenario** by filling the next column in the **control file** (**Figure 27**). The easiest is copying the first scenario (column C), pasting it (column D), and then changing the name and the relevant parameter(s).

👉 **Link** the solar and wind capacity factor rows in the **control file** → sheet "*Hydropower plant parameters*" (filas 14 y 15) **to the relevant worksheet** where you will have filled in the time series in the ***data_CF_solar.xlsx*** and ***data_CF_wind.xlsx*** files.

👉 In the **control file**, the header "*Setting solar/wind mix*" in row 19 contains the parameter *c_solar_relative*. It reflects the share of solar in the VRE mix that the hydropower plant will support; the (implied) share of wind *c_wind_relative* is then equal to *1 – c_solar_relative*. **Leaving this field empty gives a scenario without solar and wind, as in the previous Task.** In this example, set *c_solar_relative* = 0.5, meaning that REVUB will simulate a solar/wind distribution where both resources have **equal installed capacity**.

👉 Deactivate the dispatch run from the previous task by setting *HPP_active* = 0 in column C, and activate this new scenario by setting *HPP_active* = 1 in column D, as in **Figure 27**.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***King Floribert HPP with VRE*** | ***[name 3]*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 0 | 1 | 0 |
| 8 | **Importing temporal data** | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in **m^3/s**; hours in rows, years in | King Floribert | King Floribert | |
| 10 | **HPP_name_data_lateral_flow** | ***[leave empty if unsure - only in case of cascade calculations]*** name of corresponding worksheet in the "data" Excel sheets (use the "inflow" workbook) from which to pull data on lateral inflow to a downstream cascade plant (hourly flow in **m^3/s**; hours in rows, years in columns) | | | |
| 11 | **HPP_name_data_precipitation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (precipitation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | King Floribert | |
| 12 | **HPP_name_data_evaporation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (evaporation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | King Floribert | |
| 13 | **HPP_name_data_outflow_prescri** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data on prescribed (environmental/irrigation) outflow (hourly in **m^3/s**; hours in rows, years in columns) | King Floribert | King Floribert | |
| 14 | **HPP_name_data_CF_solar** | ***[leave empty if scenario has no solar]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly solar CF as **fraction**/percentage; hours in rows, years in columns) | | Solar near King Floribert | |
| 15 | **HPP_name_data_CF_wind** | ***[leave empty if scenario has no wind]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly wind CF as **fraction**/percentage; hours in rows, years in columns) | | Wind near King Floribert | |
| 16 | **HPP_name_data_load** | ***[leave empty in calibration run]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly load **normalised** to a mean of unity) | Flexibar | Flexibar | |
| 17 | **Importing bathymetry** | | | | |
| 18 | **HPP_name_data_bathymetry** | name of corresponding worksheet in the "data" Excel sheets from which to pull data (first column: volume in **m^3**; second: surface area in **m^2**; third: head in | Lake Floribert | Lake Floribert | |
| 19 | **Setting solar/wind mix** | | | | |
| 20 | **c_solar_relative** | ***[leave empty for scenarios without solar/wind]*** fraction of solar in the solar/wind portfolio | | 0.5 | |

*Figure 27: Creating a second scenario for dispatch with solar and wind power included.*

### 🏃 CODE RUN 🏃

Now run REVUB again for this new dispatch scenario.

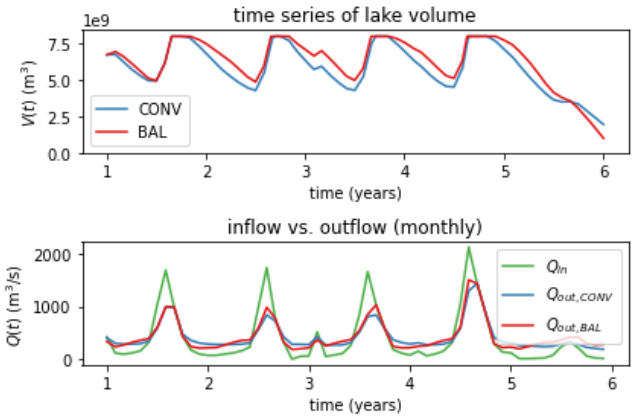
👉 When visualising results using ***C_REVUB_plotting_individual.py***, make sure to set the correct name of the scenario/plant in the **control file**, worksheet "*Plot power output (single HPP)*", as in **Figure 28**.

| | A | B | C |
|---|---|---|---|
| 1 | **parameter** | **value** | **explanation** |
| 2 | plot_HPP | King Floribert HPP with VRE | exact name of hydropower plant to plot |
| 3 | plot_year | 2 | for monthly and hourly graphs: choose which simulatior |
| 4 | plot_month | 4 | for hourly graph: choose month to plot (1 = Jan; 2 = Feb; |
| 5 | plot_day_month | 12 | for hourly graph: choose day of the month that the plot |
| 6 | plot_num_days | 3 | for hourly graph: number of days to show, starting at *pl* |
| 7 | plot_RoR_part | *1* | set to 1 to plot run-of-river component of electricity gen |
| 8 | plot_ELCC_line | *1* | set to 1 to include a bold black line for ELCC, or to 0 to le |

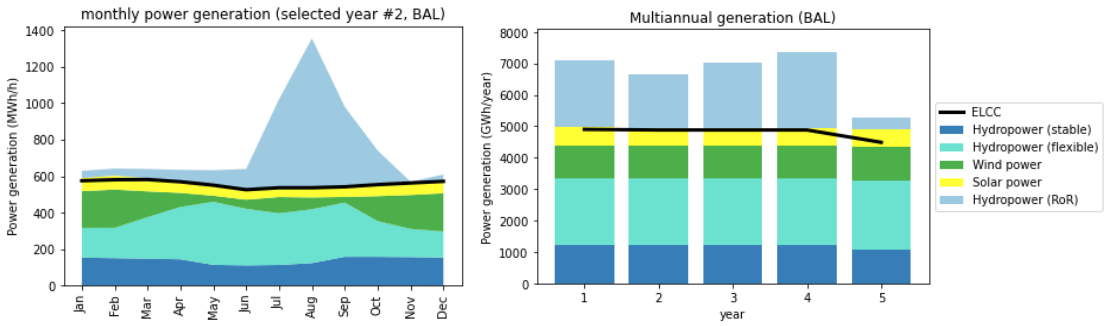
*Figure 28: Setting the correct scenario/plant name in the **control file**, worksheet "Plot power output (single HPP)".*

🎾 Examine the **implications of solar/wind integration** on lake level and power generation outcomes.

We see that the rule curve is followed well, but lake levels remain somewhat higher during the first half of each year (before the river discharge peak).




The **ELCC is boosted** by the presence of solar and wind, and total generation can be increased by about 40% in normal years thanks to solar/wind integration, as compared to the results from the dispatch run without VRE.

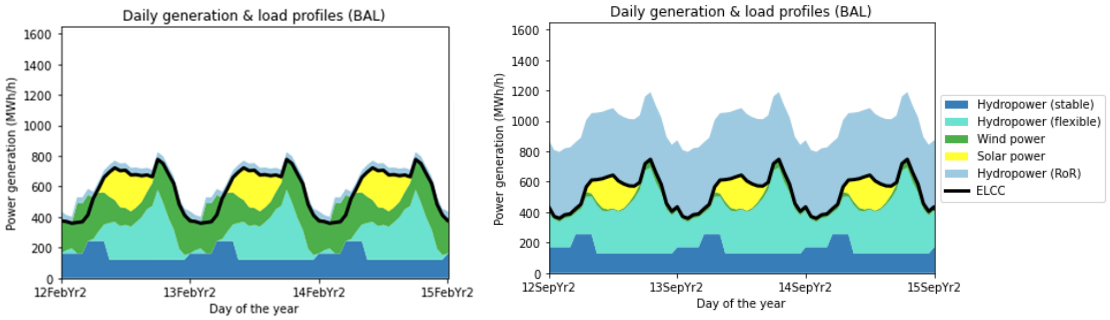



The hydropower operation changes at seasonal and sub-daily level compared to the dispatch run without VRE. In some seasons (e.g. February) solar and wind have a much higher share of the dispatched mix than in others (e.g. September).

- Hydropower still ramps up in the mornings and evenings to respond to peak loads**, but can ramp down more strongly during daytime** in all seasons, thanks to the solar PV infeed.

- In the high-wind season (e.g. February), the wind, **blowing stronger at night than during daytime**, alleviates the need for hydropower peaking in the morning.

- In some moments in February, the hydropower plant even **ramps down to its minimum stable levels** (determined by minimum turbine load and by irrigation needs). In September, when wind is very weak, this does not occur.




🎾 How many **MW of solar and wind could be hybridised** with the hydropower plant in this simulation? The answer is given by the parameter *c_multiplier_BAL*, part of the standard REVUB output.

In this case, it is about 656 MW (verify!), of which half is solar and half is wind (as *c_solar_relative* = 0.5).

# Part 3: Modelling a hydropower cascade

The Planet Bank has come up with an additional assignment. According to a group of engineers from the Republic of Flexibar, there is an **excellent site just downstream** of the King Floribert Gorge where the river undergoes a 300-meter drop across a few kilometres.

They are proposing that **an 1800-MW power plant could be built here**, and have suggested the name Princess Mariana Hydropower Plant. However, the site is **not suitable for a large reservoir**. The Princess Mariana Hydropower Plant could only have a small reservoir, capable of storing one-tenth of the volume of King Floribert Lake.

The Planet Bank wants to investigate the option of a **cascade of two hydropower plants**: King Floribert upstream as planned, and Princess Mariana downstream. This way, the operation of King Floribert Lake could be optimised to serve the Princess Mariana plant, which could effectively use the combined storage of both reservoirs.

In this section, we will investigate how to **model such a cascade** in REVUB.

### Task 3.1: Types of cascades

To couple two hydropower plants in a cascade, the **control file** → worksheet "*Hydropower plant parameters*" contains parameters that can be used to specify the coupling between the two (cf. **Figure 29**), under the header "*Defining cascades*" (row 4).

First, the user needs to decide which power plant is the "leading" unit in the cascade, and which the "following" unit.

Is the downstream unit using the combined storage capacities of both plants, with **joint reservoir operation** optimised to mainly benefit the downstream plant? In this case, the **downstream unit is leading** and the **upstream unit is following ("lagging").** The REVUB code must force the upstream plant to run according to a scheme that benefits the downstream plant.

In that case, the parameter "*HPP_cascade_upstream*" (row 5) specifies the name of the following unit for any leading unit. So, if **Princess Mariana** were to be the leading unit and **King Floribert** the lagging unit, we would specify *HPP_cascade_upstream* for the Princess Mariana plant to be "King Floribert", as in **Figure 29**.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | |
| 2 | HPP_name | name of hydropower plant in simulation | *King Floribert HPP* | *King Floribert HPP with VRE* | *King Floribert in cascade* | *Princess Mariana HPP* |
| 3 | HPP_active | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 |
| 4 | **Defining cascades** | ***[Note: for advanced REVUB users; leave empty if unsure ]*** | | | | |
| 5 | HPP_cascade_upstream | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant upstream of this one whose reservoir serves both. The upstream one must be present in this sheet, but can be set to HPP_active = 0. If it is set to HPP_active = 1, it will be forced to provide the outflow necessary to run the downstream one. | | | | King Floribert in cascade |
| 6 | f_cascade_upstream | ***[leave empty if unsure - default proportional to storage sizes will be used]*** In case this HPP is the principal flexibility provider in a cascade, and there is a hydropower plant upstream with reservoir that serves both plants, insert here the share of volume changes allocated to the upper reservoir. | | | | |
| 7 | HPP_cascade_downstream | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant downstream of this one whose outflow it turbines. If the downstream one is present in this sheet and set to HPP_active = 1, it will be run with the outflow of this HPP as input data. | | | | |

*Figure 29: Specifying cascade setups where the downstream unit is leading.*

> 💡 **TIP #3** 💡
>
> The parameter "*f_cascade_upstream*" (row 6) is optional. When the downstream unit is leading, REVUB normally assumes that potential volume fluctuations are proportionally divided across upstream and downstream reservoirs depending on their relative volume.
>
> This can be overruled by the user with the *f_cascade_upstream* parameter, representing how much of the needed volume fluctuation should be buffered by the upstream reservoir.

Is the downstream unit simply receiving the turbined flow from the upstream unit, which optimised its own reservoir operation? In this case, the **upstream unit is leading** and the **downstream unit is following**. The REVUB code must use the outflow of the upstream plant as inflow of the downstream plant.

In that case, the parameter "*HPP_cascade_downstream*" (row 7) specifies the name of the following unit for any leading unit. So, if **King Floribert** were to be the leading unit, and **Princess Mariana** the lagging unit, we would specify *HPP_cascade_downstream* for the King Floribert unit to be "Princess Mariana".

### Task 3.2: Setting up a cascade dispatch run

We will investigate the case where Princess Mariana is the **leading unit**, requiring a co-optimisation of storage upstream and downstream.

👉 Therefore, please enter two new entries (***King Floribert in cascade***, and ***Princess Mariana HPP***) in the **control file**, and link them together exactly as shown in **Figure 29**.

Which temporal data (as in **Task 1.2**) do we link to the cascaded plants?

👉 We assume there is negligible lateral inflow between the two plants, so we can **use the same inflow time series** as before. We also assume the same precipitation and evaporation patterns. Thus, the parameters "*HPP_name_data_inflow*", "*HPP_name_data_precipitation*", and "*HPP_name_data_evaporation*" will be equal to those we used earlier in the simulations of King Floribert Hydropower Plant.

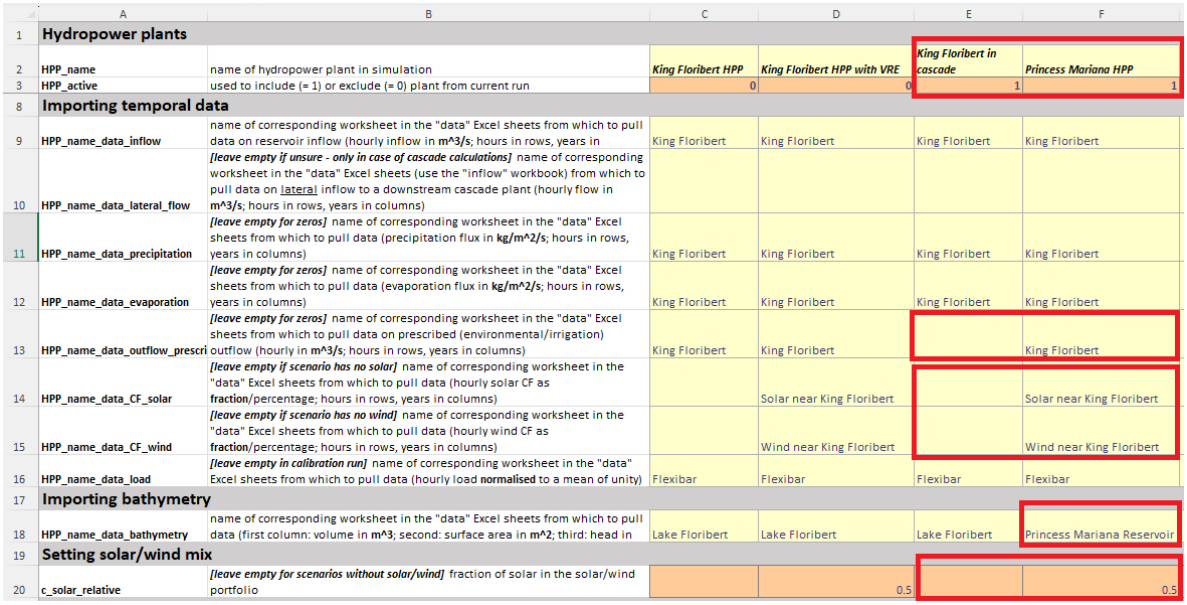
👉 The **planned irrigation schemes are assumed to be all downstream** of the Princess Mariana location. Therefore, we will link the prescribed outflow file to the column containing ***Princess Mariana***, and can delete it from the column containing ***King Floribert in cascade***, as in **Figure 30**.

👉 Assume that the **leading unit will be responsible for providing the flexibility** needed for solar and wind support. Assume that the lagging unit will not have to do this. Therefore, link the solar and wind profiles to ***Princess Mariana HPP*** and remove them from ***King Floribert in cascade***, as in **Figure 30**.

👉 For the bathymetry of the small future Princess Mariana reservoir, please use the data given in the Excel file ***raw_data.xlsx*** → worksheet "*bathymetry_2*".

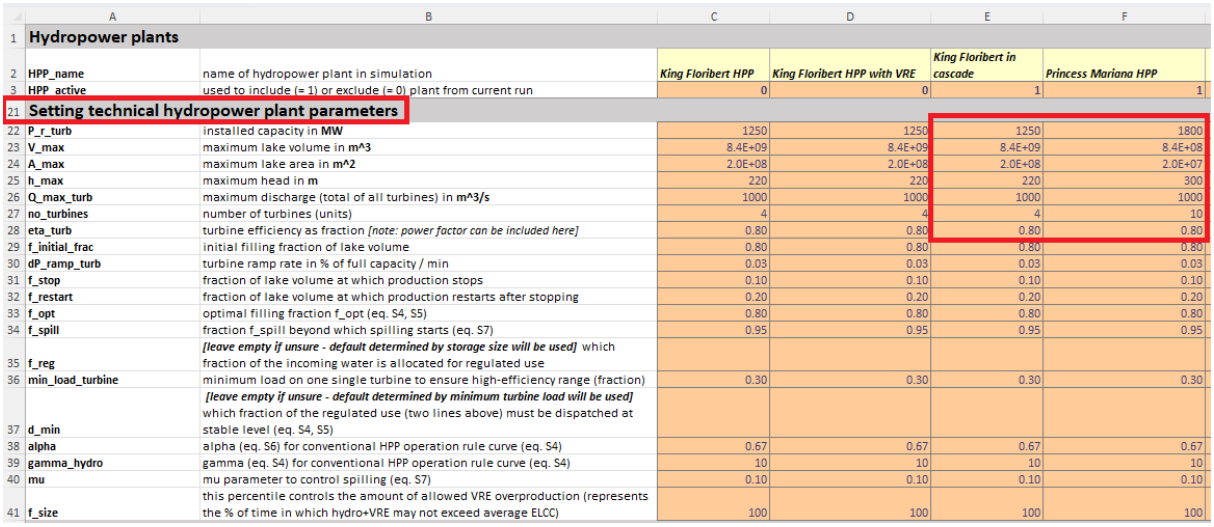
👉 Assume that all technical parameters for Princess Mariana plant are equal to those you used earlier for King Floribert, except for the ones indicated under the header "Princess Mariana" in the file ***raw_data.xlsx*** → worksheet "*other technical parameters_2*". The **control file** should look as in **Figure 31**.

👉 We keep the calibration period at 1980-1984.



| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***King Floribert HPP with VRE*** | ***King Floribert in cascade*** | ***Princess Mariana HPP*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 |
| 8 | **Importing temporal data** | | | | | |
| 9 | **HPP_name_data_inflow** | name of corresponding worksheet in the "data" Excel sheets from which to pull data on reservoir inflow (hourly inflow in **m^3/s**; hours in rows, years in | King Floribert | King Floribert | King Floribert | King Floribert |
| 10 | **HPP_name_data_lateral_flow** | ***[leave empty if unsure - only in case of cascade calculations]*** name of corresponding worksheet in the "data" Excel sheets (use the "inflow" workbook) from which to pull data on lateral inflow to a downstream cascade plant (hourly flow in **m^3/s**; hours in rows, years in columns) | | | | |
| 11 | **HPP_name_data_precipitation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (precipitation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | King Floribert | King Floribert | King Floribert |
| 12 | **HPP_name_data_evaporation** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (evaporation flux in **kg/m^2/s**; hours in rows, years in columns) | King Floribert | King Floribert | King Floribert | King Floribert |
| 13 | **HPP_name_data_outflow_prescri** | ***[leave empty for zeros]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data on prescribed (environmental/irrigation) outflow (hourly in **m^3/s**; hours in rows, years in columns) | King Floribert | King Floribert | | King Floribert |
| 14 | **HPP_name_data_CF_solar** | ***[leave empty if scenario has no solar]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly solar CF as **fraction**/percentage; hours in rows, years in columns) | | Solar near King Floribert | | Solar near King Floribert |
| 15 | **HPP_name_data_CF_wind** | ***[leave empty if scenario has no wind]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly wind CF as **fraction**/percentage; hours in rows, years in columns) | | Wind near King Floribert | | Wind near King Floribert |
| 16 | **HPP_name_data_load** | ***[leave empty in calibration run]*** name of corresponding worksheet in the "data" Excel sheets from which to pull data (hourly load **normalised** to a mean of unity) | Flexibar | Flexibar | Flexibar | Flexibar |
| 17 | **Importing bathymetry** | | | | | |
| 18 | **HPP_name_data_bathymetry** | name of corresponding worksheet in the "data" Excel sheets from which to pull data (first column: volume in **m^3**; second: surface area in **m^2**; third: head in | Lake Floribert | Lake Floribert | Lake Floribert | Princess Mariana Reservoir |
| 19 | **Setting solar/wind mix** | | | | | |
| 20 | **c_solar_relative** | ***[leave empty for scenarios without solar/wind]*** fraction of solar in the solar/wind portfolio | | 0.5 | | 0.5 |

*Figure 30: Specifying input time series for two cascaded plants.*



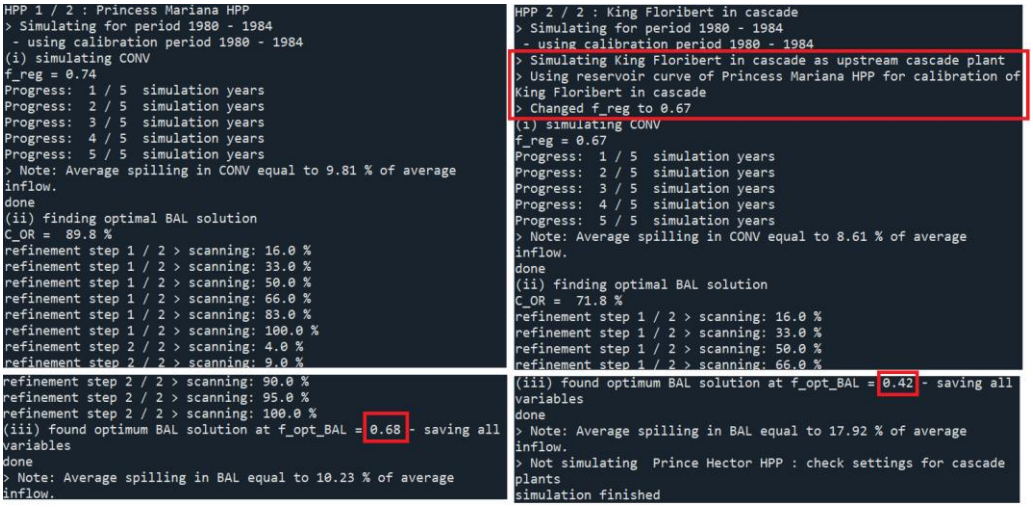
| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | ***King Floribert HPP*** | ***King Floribert HPP with VRE*** | ***King Floribert in cascade*** | ***Princess Mariana HPP*** |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 |
| 21 | **Setting technical hydropower plant parameters** | | | | | |
| 22 | **P_r_turb** | installed capacity in **MW** | 1250 | 1250 | 1250 | 1800 |
| 23 | **V_max** | maximum lake volume in **m^3** | 8.4E+09 | 8.4E+09 | 8.4E+09 | 8.4E+08 |
| 24 | **A_max** | maximum lake area in **m^2** | 2.0E+08 | 2.0E+08 | 2.0E+08 | 2.0E+07 |
| 25 | **h_max** | maximum head in **m** | 220 | 220 | 220 | 300 |
| 26 | **Q_max_turb** | maximum discharge (total of all turbines) in **m^3/s** | 1000 | 1000 | 1000 | 1000 |
| 27 | **no_turbines** | number of turbines (units) | 4 | 4 | 4 | 10 |
| 28 | **eta_turb** | turbine efficiency as fraction *[note: power factor can be included here]* | 0.80 | 0.80 | 0.80 | 0.80 |
| 29 | **f_initial_frac** | initial filling fraction of lake volume | 0.80 | 0.80 | 0.80 | 0.80 |
| 30 | **dP_ramp_turb** | turbine ramp rate in % of full capacity / min | 0.03 | 0.03 | 0.03 | 0.03 |
| 31 | **f_stop** | fraction of lake volume at which production stops | 0.10 | 0.10 | 0.10 | 0.10 |
| 32 | **f_restart** | fraction of lake volume at which production restarts after stopping | 0.20 | 0.20 | 0.20 | 0.20 |
| 33 | **f_opt** | optimal filling fraction f_opt (eq. S4, S5) | 0.80 | 0.80 | 0.80 | 0.80 |
| 34 | **f_spill** | fraction f_spill beyond which spilling starts (eq. S7) | 0.95 | 0.95 | 0.95 | 0.95 |
| 35 | **f_reg** | ***[leave empty if unsure - default determined by storage size will be used]*** which fraction of the incoming water is allocated for regulated use | | | | |
| 36 | **min_load_turbine** | minimum load on one single turbine to ensure high-efficiency range (fraction) | 0.30 | 0.30 | 0.30 | 0.30 |
| 37 | **d_min** | ***[leave empty if unsure - default determined by minimum turbine load will be used]*** which fraction of the regulated use (two lines above) must be dispatched at stable level (eq. S4, S5) | | | | |
| 38 | **alpha** | alpha (eq. S6) for conventional HPP operation rule curve (eq. S4) | 0.67 | 0.67 | 0.67 | 0.67 |
| 39 | **gamma_hydro** | gamma (eq. S4) for conventional HPP operation rule curve (eq. S4) | 10 | 10 | 10 | 10 |
| 40 | **mu** | mu parameter to control spilling (eq. S7) | 0.10 | 0.10 | 0.10 | 0.10 |
| 41 | **f_size** | this percentile controls the amount of allowed VRE overproduction (represents the % of time in which hydro+VRE may not exceed average ELCC) | 100 | 100 | 100 | 100 |

*Figure 31: Differentiating technical parameters between two cascade plants.*

### 🏃 CODE RUN 🏃

Now run all three REVUB scripts, as before (***A_REVUB_initialise.py***, ***B_REVUB_main_code.py*** and ***C_REVUB_plotting_individual.py***). Activate the two cascade plants by setting *HPP_active* = 1.

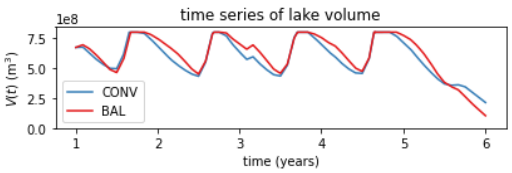
When running REVUB for the cascade, you will see that the code **automatically runs the leading unit first** (Princess Mariana), regardless of the order in which you entered them in the control file.



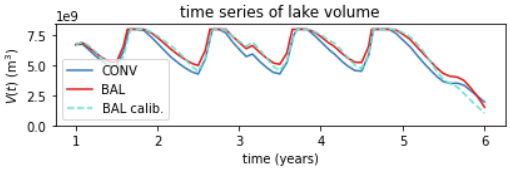

The lagging unit is run after the leading unit. To force the lagging unit to align operation with what the leading unit needs, the code uses the **reservoir drawdown-refill needs** implied by the **simulation of the leading unit using cumulative reservoir storage** (the sum of storage in both reservoirs) as reference for the lagging unit's dispatch—instead of the rule curves from **Task 2.1**. Thus, those rule curves are overruled in the cascade setup.

🔎 Examine the **implications of constructing the cascade** on lake level and power generation outcomes.

The results for **stored volume** are shown below (**left panel** for the **leading/downstream** unit; **right panel** for the **lagging/upstream** unit). In this case, a curve "*BAL calib*", representing the new rule curve for the lagging unit, has been added.




***Princess Mariana (leading unit)***




***King Floribert (lagging unit)***

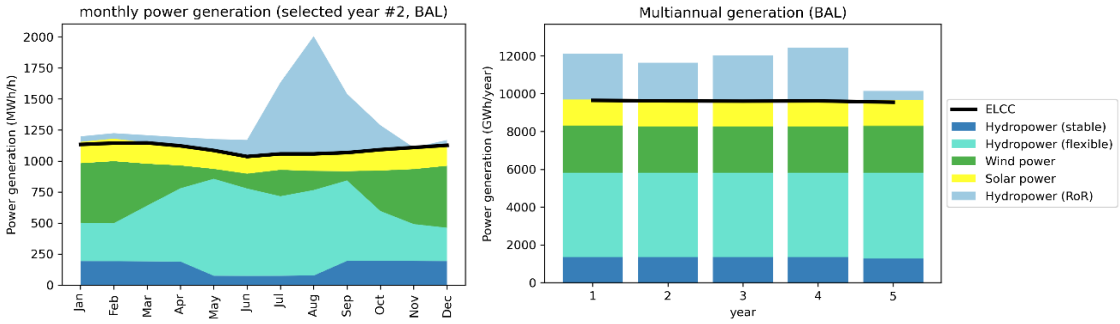
Examining the seasonal and hourly dispatch patterns from the leading/downstream unit Princess Mariana, we see that a substantially higher solar/wind hybridisation is possible (*c_multiplier_BAL* = 1562 MW), as compared to the non-cascade simulations of King Floribert only (compare to the results in **Task 2.4**). The graphics further below indicate some of the results obtained for the Princess Mariana plant.

This increase is mainly thanks to three reasons:

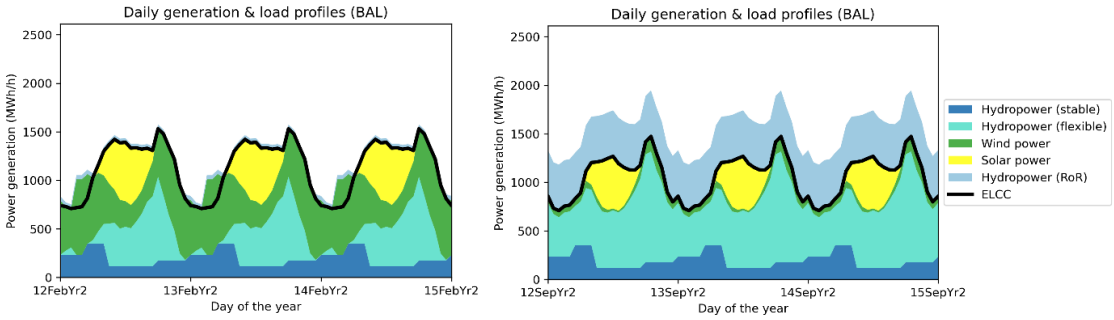
- The Princess Mariana plant has its **capacity divided over more turbines** (**Figure 31**), hence lower minimum load than King Floribert (cf. the ratio of the *stable* and *flexible* hydropower component);

- The Princess Mariana reservoir volume adds one-tenth in **regulation capacity** to the total storage volume in the cascade, adding to the storage upstream;

- Princess Mariana hydropower plant has **less intense lake level fluctuations** because of its different bathymetry, and can sustain more consistent levels of power output.




In the daily dispatch profiles, we can see this reflected e.g. in the lower minimum stable load (compared to the non-cascade case from **Task 2.4**), which provides more space to increase solar-wind hybridisation.




### Task 3.3: Adding a run-of-river plant

The same group of engineers from Flexibar say that there is **one more site for hydropower development** on the same river, downstream of the Princess Mariana site.

They argue that a 500-MW hydropower plant could be built here. This could be a **run-of-river scheme**, designed to provide baseload by **turbining the regulated outflow** from the upstream reservoirs. The proposed name of the plant is Prince Hector.

This would make it a **three-plant cascade**: King Floribert (upstream), followed by Princess Mariana (in the middle), followed by Prince Hector (downstream).

In the logic of "leading" and "lagging" plants in a cascade, Prince Hector is clearly "lagging", turbining what it receives from the Princess Mariana plant upstream.

👉🏽 Add the run-of-river Prince Hector plant to the **control file** and set its link with the cascade correctly, by specifying the parameter *HPP_cascade_downstream* for the Princess Mariana unit to be "Prince Hector", as in **Figure 32**.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | | |
| 2 | HPP_name | name of hydropower plant in simulation | *King Floribert HPP* | *King Floribert HPP with VRE* | *King Floribert in cascade* | *Princess Mariana HPP* | *Prince Hector HPP* |
| 3 | HPP_active | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 | 1 |
| 4 | **Defining cascades [*Note: for advanced REVUB users; leave empty if unsure*]** | | | | | | |
| 5 | HPP_cascade_upstream | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant upstream of this one whose reservoir serves both. The upstream one must be present in this sheet, but can be set to HPP_active = 0. If it is set to HPP_active = 1, it will be forced to provide the outflow necessary to run the downstream one. | | | | King Floribert in cascade | |
| 6 | f_cascade_upstream | ***[leave empty if unsure - default proportional to storage sizes will be used]*** In case this HPP is the principal flexibility provider in a cascade, and there is a hydropower plant upstream with reservoir that serves both plants, insert here the share of volume changes allocated to the upper reservoir. | | | | | |
| 7 | HPP_cascade_downstream | In case this HPP is the principal flexibility provider in a cascade, insert here (if applicable) the name of the hydropower plant downstream of this one whose outflow it turbines. If the downstream one is present in this sheet and set to HPP_active = 1, it will be run with the outflow of this HPP as input data. | | | | Prince Hector HPP | |

*Figure 32: Specifying cascade setups where the downstream unit is lagging.*

👉 As the Prince Hector plant is run-of-river, we only need a very reduced set of technical data. For instance, bathymetry data is not necessary, since there is no reservoir. Use the technical parameters indicated in the file ***raw_data.xlsx*** → worksheet "*other technical parameters_2*", under the header "*Prince Hector*". **Leave all other data points empty.** The **control file** should look as in **Figure 33**.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | *King Floribert HPP* | *King Floribert HPP with VRE* | *King Floribert in cascade* | *Princess Mariana HPP* | *Prince Hector HPP* |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 | 1 |
| 21 | **Setting technical hydropower plant parameters** | | | | | | |
| 22 | **P_r_turb** | installed capacity in **MW** | 1250 | 1250 | 1250 | 1800 | 500 |
| 23 | **V_max** | maximum lake volume in **m^3** | 8.4E+09 | 8.4E+09 | 8.4E+09 | 8.4E+08 | |
| 24 | **A_max** | maximum lake area in **m^2** | 2.0E+08 | 2.0E+08 | 2.0E+08 | 2.0E+07 | |
| 25 | **h_max** | maximum head in **m** | 220 | 220 | 220 | 300 | 100 |
| 26 | **Q_max_turb** | maximum discharge (total of all turbines) in **m^3/s** | 1000 | 1000 | 1000 | 1000 | 600 |
| 27 | **no_turbines** | number of turbines (units) | 4 | 4 | 4 | 10 | 3 |
| 28 | **eta_turb** | turbine efficiency as fraction *[note: power factor can be included here]* | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 |
| 29 | **f_initial_frac** | initial filling fraction of lake volume | 0.80 | 0.80 | 0.80 | 0.80 | |
| 30 | **dP_ramp_turb** | turbine ramp rate in % of full capacity / min | 0.03 | 0.03 | 0.03 | 0.03 | |
| 31 | **f_stop** | fraction of lake volume at which production stops | 0.10 | 0.10 | 0.10 | 0.10 | |
| 32 | **f_restart** | fraction of lake volume at which production restarts after stopping | 0.20 | 0.20 | 0.20 | 0.20 | |
| 33 | **f_opt** | optimal filling fraction f_opt (eq. S4, S5) | 0.80 | 0.80 | 0.80 | 0.80 | |
| 34 | **f_spill** | fraction f_spill beyond which spilling starts (eq. S7) | 0.95 | 0.95 | 0.95 | 0.95 | |
| 35 | **f_reg** | ***[leave empty if unsure - default determined by storage size will be used]*** which fraction of the incoming water is allocated for regulated use | | | | | |
| 36 | **min_load_turbine** | minimum load on one single turbine to ensure high-efficiency range (fraction) | 0.30 | 0.30 | 0.30 | 0.30 | |
| 37 | **d_min** | ***[leave empty if unsure - default determined by minimum turbine load will be used]*** which fraction of the regulated use (two lines above) must be dispatched at stable level (eq. S4, S5) | | | | | |
| 38 | **alpha** | alpha (eq. S6) for conventional HPP operation rule curve (eq. S4) | 0.67 | 0.67 | 0.67 | 0.67 | |
| 39 | **gamma_hydro** | gamma (eq. S4) for conventional HPP operation rule curve (eq. S4) | 10 | 10 | 10 | 10 | |
| 40 | **mu** | mu parameter to control spilling (eq. S7) | 0.10 | 0.10 | 0.10 | 0.10 | |
| 41 | **f_size** | this percentile controls the amount of allowed VRE overproduction (represents the % of time in which hydro+VRE may not exceed average ELCC) | 100 | 100 | 100 | 100 | |

*Figure 33: Specifying technical parameters for a run-of-river plant*

👉 We already know the optimal operation that the model will decide for the upstream plants (the Prince Hector plant has no influence on this, as it is "lagging"). Therefore, to speed up the simulation, we can use the advice from **TIP #2** in **Task 2.3**, using the previously identified optimal parameterisation that REVUB indicated in the console during the code execution. This would look as in **Figure 34** in the **control file**, using the parameter *f_init_BAL_opt*.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | **Hydropower plants** | | | | | | |
| 2 | **HPP_name** | name of hydropower plant in simulation | *King Floribert HPP* | *King Floribert HPP with VRE* | *King Floribert in cascade* | *Princess Mariana HPP* | *Prince Hector HPP* |
| 3 | **HPP_active** | used to include (= 1) or exclude (= 0) plant from current run | 0 | 0 | 1 | 1 | 1 |
| 48 | **Setting simulation accuracy** | | | | | | |
| 49 | **f_init_BAL_start** | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the lower value in the guessed range. If not sure, start with 0. | 0 | 0 | 0 | 0 | |
| 50 | **f_init_BAL_step** | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the step size in the guessed range. If not sure, start with a decimal fraction of unity, e.g. 0.1. | 0.2 | 0.2 | 0.2 | 0.2 | |
| 51 | **f_init_BAL_end** | These values are used to get a good initial guess for the range of the ELCC for BAL. This is the upper value in the guessed range. If not sure, start with 1. | 1 | 1 | 1 | 1 | |
| 52 | **N_refine_BAL** | Number of refinement loops for equilibrium search for min(Psi) in BAL (see eq. S21). Usually, 2 suffices; use 1 for quicker (but less accurate) simulations. | 2 | 2 | 2 | 2 | |
| 53 | **f_init_BAL_opt** | ***[only use if already known from previous run]*** The optimal value in the range of the ELCC for BAL | | | 0.42 | 0.68 | |

*Figure 34: Pre-setting the optimisation parameters at their identified values to speed up simulations*

### 🏃 CODE RUN 🏃

Now run all three REVUB scripts, as before (***A_REVUB_initialise.py***, ***B_REVUB_main_code.py*** and ***C_REVUB_plotting_individual.py***). Activate the three cascade plants by setting *HPP_active* = 1, as in **Figure 33**.

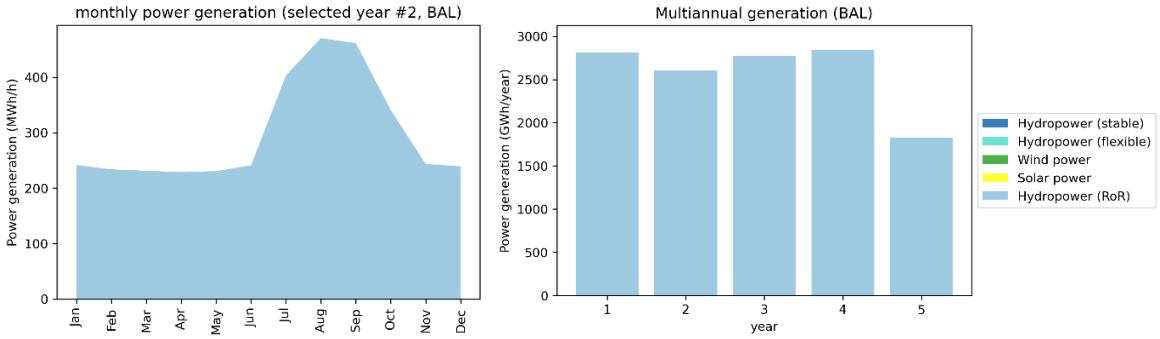
The output on the console will look as below for the three units in the cascade. It is indicated which outflow of an upstream plant is used as inflow to the downstream run-of-river plant.

```
HPP 1 / 3 : Princess Mariana HPP
> Simulating for period 1980 - 1984
 - using calibration period 1980 - 1984
(i) simulating CONV
f_reg = 0.74
Progress:  1 / 5  simulation years
Progress:  2 / 5  simulation years
Progress:  3 / 5  simulation years
Progress:  4 / 5  simulation years
Progress:  5 / 5  simulation years
> Note: Average spilling in CONV equal to 9.81 % of average
inflow.
done
(ii) finding optimal BAL solution
C_OR =  89.8 %
skipping optimisation - using user-specified optimum directly
(iii) found optimum BAL solution at f_opt_BAL = 0.68 - saving
all variables
done
> Note: Average spilling in BAL equal to 10.23 % of average
inflow.
```

```
HPP 2 / 3 : King Floribert in cascade
> Simulating for period 1980 - 1984
 - using calibration period 1980 - 1984
> Simulating King Floribert in cascade as upstream cascade plant
> Using reservoir curve of Princess Mariana HPP for calibration of
King Floribert in cascade
> Changed f_reg to 0.67
(i) simulating CONV
f_reg = 0.67
Progress:  1 / 5  simulation years
Progress:  2 / 5  simulation years
Progress:  3 / 5  simulation years
Progress:  4 / 5  simulation years
Progress:  5 / 5  simulation years
> Note: Average spilling in CONV equal to 8.61 % of average
inflow.
done
(ii) finding optimal BAL solution
C_OR =  71.8 %
skipping optimisation - using user-specified optimum directly
(iii) found optimum BAL solution at f_opt_BAL = 0.42 - saving all
variables
done
> Note: Average spilling in BAL equal to 17.92 % of average
inflow.
```

```
HPP 3 / 3 : Prince Hector HPP
> Simulating for period 1980 - 1984
> Simulating Prince Hector HPP as downstream cascade plant
> Using outflow of Princess Mariana HPP as inflow of Prince Hector
HPP
> Simulating Prince Hector HPP as run-of-river plant
simulation finished
```

🎾 Verify that the seasonal and yearly production curves from the Prince Hector hydropower plant look as below after simulation. The production from Prince Hector is **determined by the outflow** from the Princess Mariana reservoir, which in turn is **determined by the joint operation** of the King Floribert and Princess Mariana plants.



# Part 4: Simulating the power mix (thermal, hydro, VRE)

Now that we know how the power generation characteristics of the cascade could look, the Planet Bank wants to know what the effects would be on the Republic of Flexibar's overall power mix.

In particular, it wants to investigate how the construction of the cascade, including hydro-supported VRE, would **reduce the fuel imports** needed to run the aging thermal plants of the Republic.

### Task 4.1: Investigating power mix composition under static demand

Currently, the Republic has **5 GW of installed thermal capacity**, and an average annual demand of 26.3 TWh, which translates to an **average load of 3.0 GW** [26.3 TWh / (365 days * 24 hours) = 3.0 GW].

The REVUB code allows to investigate **how the thermal capacity would behave** in the presence of hydro and VRE, assuming that the latter have **priority of dispatch**.

To do this, first, in the the **control file** → sheet "*Plot power output (multi HPP)*", we can enter a set of hydropower plants whose production time series we want to sum up. We can also enter parameters on the other available capacity (supposed to be thermal, and/or for importing) in a country's power mix.

Given the above data, we can enter the needed data in the **control file** → sheet "*Plot power output (multi HPP)*" (**Figure 35**).

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **parameter** | **value(s)** | | | | | | | | |
| 2 | | *enter the names of the hydropower plants to include in this plot in row 2, starting in column B* | | | | | | | | |
| 3 | plot_HPP_multiple | King Floribert in cascade | Princess Mariana HPP | Prince Hector HPP | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | *[choose] whether to use static (=0; same year-to-year) or dynamic (=1; time series) demand and capacity* | | | | | | | | |
| 6 | choose_demand_type | *0* | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | *[static: demand]* | | | | | | | | |
| 9 | P_total_av | 3000 | total average load (MW) assumed for graph of full power mix | | | | | | | |
| 10 | | | | | | | | | | |
| 11 | | *[static: non-hydro capacity]* | | | | | | | | |
| 12 | P_r_total_thermal | 5000 | total flexible non-hydro (thermal, importc, &c.) capacity assumed available (MW) | | | | | | | |
| 13 | | | | | | | | | | |
| 14 | | *[dynamic: demand] enter a time series of average demand (MW) for each year in the simulation* | | | | | | | | |
| 15 | | *[year 1]* | *[year 2]* | *&c...* | | | | | | |
| 16 | P_total_av_series | | | | | | | | | |
| 17 | | | | | | | | | | |
| 18 | | *[dynamic: non-hydro capacity] enter a time series of available capacity (MW) of flexible non-hydro assets (thermal, imports, &c.)* | | | | | | | | |
| 19 | | *[year 1]* | *[year 2]* | *&c...* | | | | | | |
| 20 | P_r_total_thermal_series | | | | | | | | | |
| 21 | | | | | | | | | | |
| 22 | | *[plotting] set visualisation parameters* | | | | | | | | |
| 23 | plot_year_multiple | 2 | for monthly and hourly graphs: choose which simulation year to take (1 = first year; 2 = second year; etc.) | | | | | | | |
| 24 | plot_month_multiple | 4 | for hourly graph: choose month to plot (1 = Jan; 2 = Feb; etc.) | | | | | | | |
| 25 | plot_day_month_multiple | 12 | for hourly graph: choose day of the month that the plot starts with | | | | | | | |
| 26 | plot_num_days_multiple | 3 | for hourly graph: number of days to show, starting at *plot_day_month* | | | | | | | |
| 27 | plot_ramping_range | 7 | number of hours to include in ramping envelopes | | | | | | | |
| 28 | chosen_load | Flexibar | enter here the name of the target load curve to be used for the plot | | | | | | | |
| 29 | plot_ELCC_line_multiple | *0* | set to 1 to include a bold black line for ELCC, or to 0 to leave out | | | | | | | |
| 30 | tables_on_multiple | *1* | set to 1 to output key time series in Excel format | | | | | | | |

*Figure 35: Entering data for full power mix simulations in the* ***control file****, on the sheet "Plot power output (multi HPP)".*

👉 Input the names of the three connected cascade plants under the variable "*plot_HPP_multiple*".

👉 Set the parameter "*choose_demand_type*" = 0 (assume the demand is static, i.e. constant from year to year).

👉 Set the parameter "*P_total_av*" = 3000 MW (the total average demand) and the parameter "*P_r_total_thermal*" = 5000 MW (the total installed thermal capacity).

👉 Leave the parameters "*P_total_av_series*" and "*P_r_total_thermal_series*" empty for now. (If we set "*choose_demand_type*" = 1, we have to enter time series for demand and thermal capacity here, instead of the static 3000 MW and 5000 MW we use now.)

👉 Set the "*plot_year_multiple*", "*plot_month_multiple*", "*plot_day_month_multiple*", and "*plot_num_days_multiple*" parameters as per needs for the plot.

👉 There are two more parameters: "*plot_ramping_range*", which will be used to create "**ramping envelopes**" to show the speed at which the hydro and thermal plants have to increase and decrease their production in the power mix, and "*chosen_load*", which is the hourly load curve that represents the overall power demand. Use "*plot_ramping_range*" = 7 (hours), and "*chosen_load*" = *Flexibar* (the curve you used earlier in the ***data_load.xlsx*** file).

👉 Also set "*plot_ELCC_line_multiple*" = 0 to disactivate plotting the ELCC. You can set "*tables_on_multiple*" = 1 in case you wish to export results at hourly resolution.

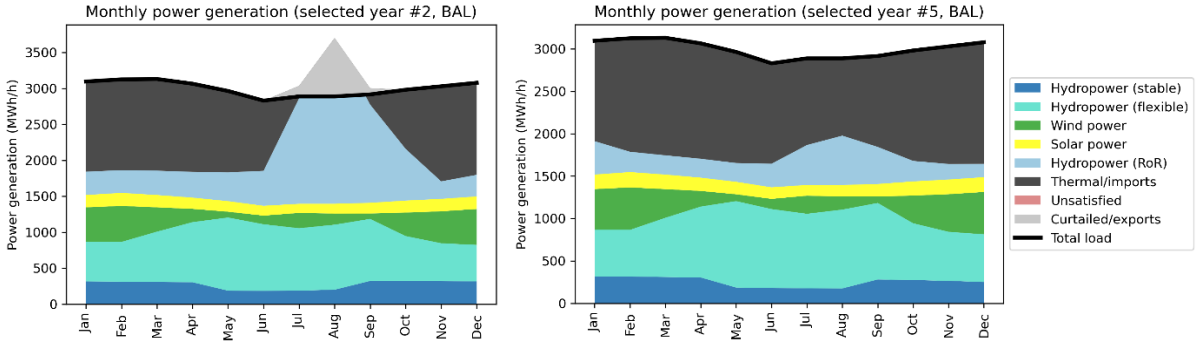
**🏃 CODE RUN 🏃**

Now run the last REVUB script, ***C_REVUB_plotting_multiple.py***. The script will produce four figures that will appear in the ***outputs*** folder, named "*Total_Fig[X]*".

🎾 Try to reproduce the below plots (by changing *plot_year_multiple*), showing **monthly power generation** for the **years #2** (normal year) and **#5** (hydrologically dry year). Here, the "hydro" statistics include all three investigated power plants: King Floribert, Princess Mariana, and Prince Hector.

- The hydrologically normal year #2 would have allowed **exporting excess hydropower** during the wet season.

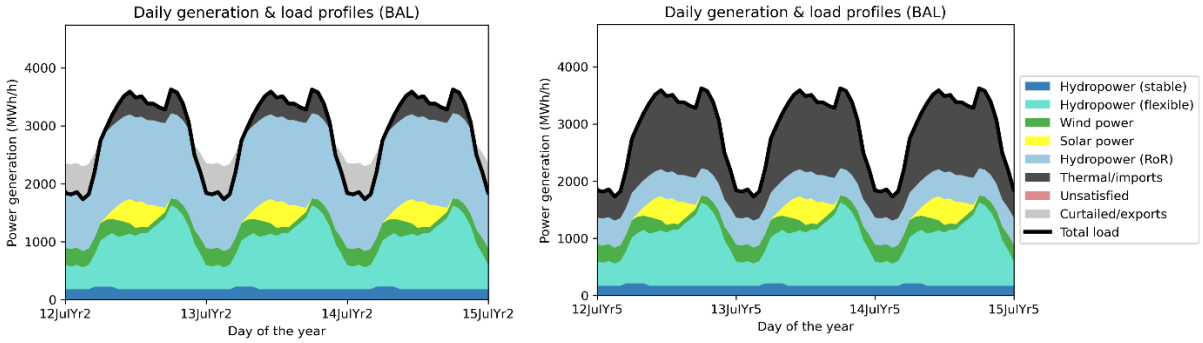
- During the hydrologically dry year #5, this would not have been possible, and **thermal power** would have had to take over.




🎾 Try to reproduce the below plots (by changing *plot_month_multiple*) to show hourly power generation statistics for the month of **July** in the **years #2** and **#5**.

- Clearly, in year #2, the Republic of Flexibar would have had **excess hydropower** during night-time in July—potentially **suitable for exports**, assuming sufficient grid capacity to neighbouring countries is available.

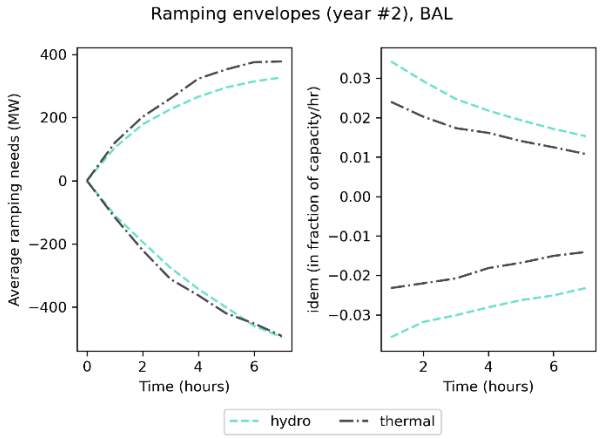
- In year #5, the dry hydrology would have prevented such a scenario, with a much stronger **baseload role for thermal power** during daytime, and insufficient hydro + VRE to even fully meet the low nighttime load.




🔎 The script also produces "**ramping envelopes**", showing the **rate of change in time** of the production of flexible hydropower and thermal power.

- The graph shows how large the **average change of power output** is across **time spans of various hours** (defined by *plot_ramping_range*).

- The graph shows how, in our scenario, hydro has **substantial responsibilities in flexibility provision**, comparable to thermal.




**Task 4.2: Investigating power mix composition under dynamic demand**

It is expected that the **demand** in the Republic of Flexibar will keep **growing by 5% per year**. At the same time, its **thermal power plant fleet is aging**, and some thermal plants need to be retired in the near future.

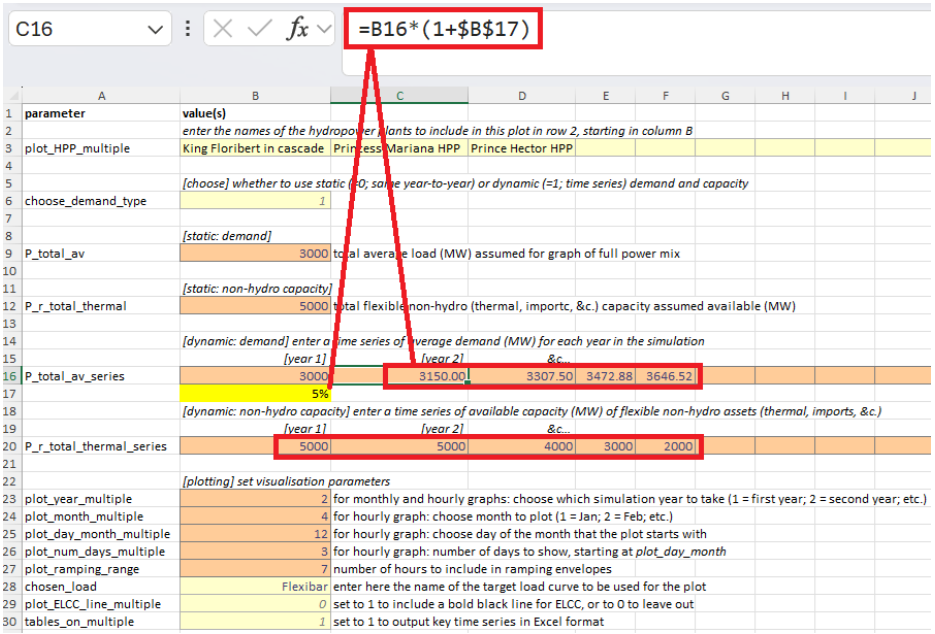
Assuming that the cascade is built, what is the chance that a future dry year (such as our hydrological year #5) will lead to **power shortages**, given the retirement of the thermal fleet?

We can investigate this through the **control file** → sheet "*Plot power output (multi HPP)*".

👉 Set the parameter "*choose_demand_type*" = 1, as in **Figure 36**.

👉 For "*P_total_av_series*", enter a time series that starts at 3000 MW in year #1 and grows with 5% every year, as in in **Figure 36**.

👉 For "*P_r_total_thermal_series*", enter a time series that starts at 5000 MW in year #1 and #2, but retiring 1000 MW in year #3, year #4, and year #5, as in **Figure 36**, to simulate the retirement of the old fleet.



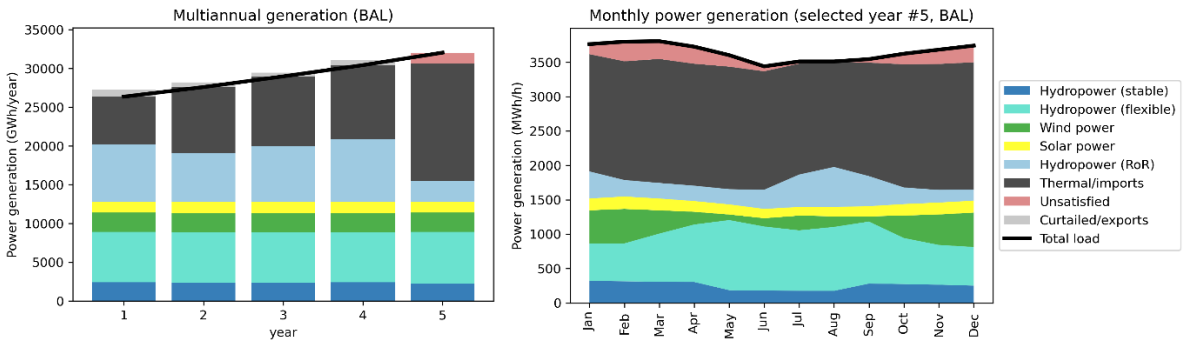

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **parameter** | **value(s)** | | | | | | | | |
| 2 | | *enter the names of the hydropower plants to include in this plot in row 2, starting in column B* | | | | | | | | |
| 3 | plot_HPP_multiple | King Floribert in cascade | Princess Mariana HPP | Prince Hector HPP | | | | | | |
| 4 | | | | | | | | | | |
| 5 | | *[choose] whether to use static (=0; same year-to-year) or dynamic (=1; time series) demand and capacity* | | | | | | | | |
| 6 | choose_demand_type | *1* | | | | | | | | |
| 7 | | | | | | | | | | |
| 8 | | *[static: demand]* | | | | | | | | |
| 9 | P_total_av | 3000 | total average load (MW) assumed for graph of full power mix | | | | | | | |
| 10 | | | | | | | | | | |
| 11 | | *[static: non-hydro capacity]* | | | | | | | | |
| 12 | P_r_total_thermal | 5000 | total flexible non-hydro (thermal, importc, &c.) capacity assumed available (MW) | | | | | | | |
| 13 | | | | | | | | | | |
| 14 | | *[dynamic: demand] enter a time series of average demand (MW) for each year in the simulation* | | | | | | | | |
| 15 | | *[year 1]* | *[year 2]* | *&c...* | | | | | | |
| 16 | P_total_av_series | 3000 | 3150.00 | 3307.50 | 3472.88 | 3646.52 | | | | |
| 17 | | 5% | | | | | | | | |
| 18 | | *[dynamic: non-hydro capacity] enter a time series of available capacity (MW) of flexible non-hydro assets (thermal, imports, &c.)* | | | | | | | | |
| 19 | | *[year 1]* | *[year 2]* | *&c...* | | | | | | |
| 20 | P_r_total_thermal_series | 5000 | 5000 | 4000 | 3000 | 2000 | | | | |
| 21 | | | | | | | | | | |
| 22 | | *[plotting] set visualisation parameters* | | | | | | | | |
| 23 | plot_year_multiple | 2 | for monthly and hourly graphs: choose which simulation year to take (1 = first year; 2 = second year; etc.) | | | | | | | |
| 24 | plot_month_multiple | 4 | for hourly graph: choose month to plot (1 = Jan; 2 = Feb; etc.) | | | | | | | |
| 25 | plot_day_month_multiple | 12 | for hourly graph: choose day of the month that the plot starts with | | | | | | | |
| 26 | plot_num_days_multiple | 3 | for hourly graph: number of days to show, starting at *plot_day_month* | | | | | | | |
| 27 | plot_ramping_range | 7 | number of hours to include in ramping envelopes | | | | | | | |
| 28 | chosen_load | Flexibar | enter here the name of the target load curve to be used for the plot | | | | | | | |
| 29 | plot_ELCC_line_multiple | *0* | set to 1 to include a bold black line for ELCC, or to 0 to leave out | | | | | | | |
| 30 | tables_on_multiple | *1* | set to 1 to output key time series in Excel format | | | | | | | |

*Figure 36: Entering dynamic time series for power mix simulations.*

> **CODE RUN**
>
> Now run again the script *C_REVUB_plotting_multiple.py*.

🎾 The results show that the installed capacity would eventually be **insufficient for a dry year**, such as our year #5.

There would be **a capacity deficit** especially at the start and end of the year, with high demand and low hydropower availability.




We could play with the time series in *P_r_total_thermal_series* to investigate how much thermal capacity would have to remain in the system to ensure robustness to dry years.

### Task 4.3: Further options for power mix investigation

We could do much more detailed investigations with the data we have.

You are encouraged to explore more scenarios and dive into more of the outputs provided by REVUB. This training dataset only showed a selection of the possibilities.

The list below provides various options and questions that could be investigated. For every option, it is indicated which would be the parameter of interest in the **control file**.

- Working with scenarios that use **a longer flow period** (parameter *year_end*) from the data provided in our file ***raw_data.xlsx***, instead of only 1980-1984 as we have done throughout the exercises. For instance, check **what happens *after* the extremely dry year #5**. How long does it take for lake levels to recuperate? How does the calibration period change this? What else can we do to lower the potential of hydropower shutdowns due to low lake levels (parameter *f_reg*)? What is the **trade-off** between **preventing hydropower shutdowns** in extreme years and **supporting more VRE** in normal years?

- Comparing scenarios with and without VRE in more detail. For instance, how is the **probability of drought impacts altered** (or not) **by the inclusion of VRE** in the hydropower management schemes?

- Comparing scenarios with different **ratios of solar and wind capacity** (parameter *c_solar_relative*). Do the different seasonalities of the resources play a role in how hydropower reacts? What are the effects across hours, seasons, and years?

- Comparing scenarios allowing **overproduction of solar and wind** compared to what the reservoirs can back-up (parameter *f_size*). Does the purposeful "overbuilding" of solar and wind power help reducing thermal power demand? Does it change the onset of the impact of droughts?

- Comparing scenarios with different **allocations of flexibility within a cascade** (parameter *f_cascade_upstream*). What are the effects on lake levels across the reservoirs in a cascade?

- Comparing scenarios assuming retrofits/modernisation of hydropower plants, for instance to achieve **lower minimum turbine loads** (parameter *min_load_turbine*). What additional flexibility could be harnessed with other types of turbines?

> 💡 **TIP #4** 💡
>
> Simulations with decades of flow data can take long when running for the first time (i.e. before knowing the optimality parameters)—depending on how generous and granular the range from *f_init_BAL_start* to *f_init_BAL_end* is set. Realistically, REVUB may need in the order of half an hour per hydropower plant. We advise to do some other tasks while the simulations run. 😊
>
> When running scenarios that potentially contain much more solar and wind—e.g. scenarios with overproduction of VRE (reducing parameter *f_size*)—the range from *f_init_BAL_start* to *f_init_BAL_end* also typically needs to expand (typically up to *f_init_BAL_end* = 2-3 for *f_size* ~ 60-70). REVUB will indicate whenever no optimality can be found in the range defined by the user.

# Conclusion

This is the end of the REVUB training. You have successfully run various scenarios of hybrid hydro-VRE operation for a hydropower cascade, and investigated the effects on the overall power mix of a country looking to replace fossil fuels with hydropower and VRE.

You have looked at aspects related to (i) reservoir operation, (ii) drought resilience, (iii) cascade coordination, (iv) flexibility needs, and (v) power system adequacy.

# Acknowledgements

This training dataset benefited from review and testing by Tinne Mast (Vrije Universiteit Brussel) and Miriam Sterl (Universiteit Utrecht).

Any comments or suggestions for improvements in the training dataset are appreciated and may be communicated to the author at sebastian.sterl@vub.be.